

Service Manual

Digital Reverb/Effects Processor

Model 224X

lexicon

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PREFACE

This service manual for the Lexicon 224X Digital Reverberator has eight sections.

Section 1, **Introduction**, contains a general description of the 224X and lists the specifications.

Section 2, **Service/Warranty**, gives instructions for periodic maintenance and describes how to return units for service and order parts. In addition, it contains the limited warranty.

Section 3, **Theory of Operation**, includes a system overview and describes how each module in the 224X works and how the modules interact with one another.

Section 4, **Performance Tests and Calibration**, lists the equipment required for performance tests and calibration and describes how to conduct these procedures.

Section 5, **Troubleshooting**, contains troubleshooting procedures, including a discussion of the power supplies and the diagnostic programs.

Section 6, **Schematics and Assembly Drawings**, contains all 224X schematic and assembly drawings.

Section 7, **Parts List**, lists the part number, quantity, description, and reference for all parts.

Section 8, **Engineering Changes**, describes modifications to the 224X and provides instructions on how to make the modifications.

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1.2 Specifications

Program Capacity	Varies depending on software version.
Storage Capacity	36 registers (nonvolatile).
Reverberation Time	Adjustable in two bands from approximately 0.6 to 70 seconds (program-dependent).
Mainframe Controls	Power on switch with LED; system reset; Left and Right input level adjustments; A, B, C, and D output level adjustments.
Frequency Response	20 Hz to 15 kHz, ± 1.5 dB 20 Hz to 12 kHz, ± 0.5 dB.
Dynamic Range*	
Reverberant mode	84 dB typical, 81 dB minimum at Reference Level, 20 Hz to 20 kHz for all reverb times from 0 to 10 seconds.
Nonreverberant mode	90 dB typical, 86 dB minimum, 20 Hz to 20 kHz noise bandwidth.
Total Harmonic Distortion (THD) and Noise*	0.04% typical, 0.07% maximum at Reference Level for all reverberation times between 0 and 35 seconds.
Interchannel Crosstalk	-55 dB at 1 kHz.
Inputs	Two, balanced and transformer-isolated; impedance: 20 kilohm; maximum level adjustable: +8 to 18 dBm.
Outputs	Four, balanced and transformer-isolated; impedance: 90 ohm; maximum level adjustable: +8 to +18 dBm; power-on muting.
Control Head Cable	15 ft standard; optional 25-ft and 50-ft cables available.
Power	Nominal: 100, 120, 220, 240 Vac (-10%, +5%) switch-selectable; 50 to 60 Hz; 150 W.
RFI Shielding	ac power connector, audio connectors, and console cable are RFI-shielded. Complies with FCC limits for Class A computing device.
Protection	Mains and secondaries fused; voltage crowbar and/or current limiting, thermal protection.

1 INTRODUCTION

1.1 Description

This service manual contains specifications, service and warranty information, theory of operation, performance tests and calibration procedures, a troubleshooting guide, schematics and assembly drawings, a parts list, and engineering changes for the 224X Reverberation/Effects Processor. This manual can be used as a reference for standard servicing procedures.

IMPORTANT: As a result of improvements and updates to the 224X, the unit being serviced may differ slightly from the descriptions and specifications in this manual. Service operations must be performed in the order described by a competent service technician only. If you have doubts about performing a procedure, please contact your Lexicon dealer or Lexicon for assistance. Lexicon is not responsible for damage resulting from incorrectly followed service procedures. Lexicon has taken considerable care in determining the accuracy of the information in this manual; however, it is not responsible for consequential damage resulting from the implementation of the procedures described.

WARNING: Hazardous voltages exist inside this unit when the power cord is connected; use extreme caution when servicing or adjusting. Service must be performed by qualified service personnel. Always place the unit on an isolation transformer before servicing.

PRECAUTIONS

Many of the internal components of this unit are extremely sensitive to static electricity. To ensure that static charges are dissipated safely, do not hand a component or board directly to another person -- place the device on a nonconductive surface and then have it picked up. The following practices minimize possible damage to ICs that can result from electrostatic discharge:

- 1 Minimize handling of integrated circuits (ICs).
- 2 Keep parts in original containers until ready for use.
- 3 Discharge personal static before handling devices.
- 4 Handle each IC by its body.
- 5 Use antistatic containers for handling and transport.
- 6 Do not slide devices over a surface.
- 7 Avoid plastic, vinyl, or styrofoam in the work area.
- 8 When removing plug-in boards, handle only by nonconductive surfaces and never touch open-edge connectors except at a static-free work station. Placing shorting strips on edge connectors usually provides complete protection to installed ICs.
- 9 Handle ICs only at a static-free work station.
- 10 Use only grounded-tip soldering irons.

Always disconnect the power cord before servicing internal components.

Connectors

Mainframe	Power: standard IEC 3 pin Audio: XLR-3 Control Head: DB-25 Optional Automation Interface: DB-25.
Control Head	Mainframe Cable: DB-25.
Serviceability	Field-serviceable, each major assembly removable.
Diagnostic Programs	Control and display via remote controller.
Cooling	Convection-cooled power supply, forced-air cooling of logic boards.
Environment	Operating: 0 to 35°C (32 to 95°F) Storage: -30 to 75°C (-22 to 167°F) Relative humidity: 95% maximum (without condensation).
Size	
Mainframe	Standard 19-in. rack mount: 19"w x 7"h x 15"d (483 x 178 x 381 mm).
Control Head	5.4"w x 8.8"h x 3"d (137.2 x 223.5 x 76.2 mm).
Weight	
Mainframe	34 lb (15.5 kg); 48 lb (22 kg) shipping.
Control Head	2.5 lb (1.2 kg); 6 lb (2.7 kg) shipping.
Automation Interface	Optional RS-232C serial interface.

Specifications subject to change without notice.

* The Reference Level is set using the zero delay-line diagnostic test program with input level adjustment set just below level at which the +12 dB LED lights with a 1-kHz tone at the input and with output sensitivity set to produce +12 dBm with a 600-ohm load. For reverberant mode, measurements are made using the Concert Hall program with the Mode Enhancement toggle off; for nonreverberant mode, measurements are made using the zero delay-line diagnostic test program.

2 SERVICE/WARRANTY

2.1 Periodic Maintenance

Under normal conditions, the 224X requires minimal maintenance. At six-month intervals, clean or replace the air filters on the right side and front panel of the mainframe. Filter elements can be cleaned using a mild detergent and warm water, and new filters can be obtained from Lexicon (front-panel filter: Lexicon no. 720-03386; side-panel filter: Lexicon no. 720-01261).

Use a soft lint-free cloth lightly dampened with a mild detergent and warm water to clean the exterior surfaces of the 224X. During servicing, use a vacuum or blower to clean dust out of the interior of the 224X.

2.2 Returning Units for Service

If the 224X must be returned to Lexicon or a designated facility for service, Lexicon assumes no responsibility for the unit in shipment from customer to factory, whether in or out of warranty. All shipments must be well packed (using the original packing materials, if possible), properly insured, and consigned to a reliable agent, such as UPS or Federal Air Express. If original packing materials are not available, please procure a new packing kit from Lexicon.

Before returning a unit, always consult with Lexicon to determine the extent of the problem and to decide on a shipping procedure.

When returning a unit for service, include the following information:

- Name
- Company name
- Address
- City, State, ZIP
- Telephone number (include area code)
- Serial number of unit
- Description of problem
- Desired return date
- Preferred method of return shipment

Please include a note describing conversations with Lexicon personnel, and give the name and telephone number of the person directly responsible for maintaining the equipment. Do NOT include accessories, such as power cords, manuals, or remote switches.

2.3 Module Exchange Program

If a defective module is clearly identified, Lexicon can usually provide a repair/exchange module within 24 hours in advance of receipt of the defective module. If a fast turnaround is required, Lexicon can ship a module by Federal Air Express or other expedited air service, resulting in

24-hour delivery if the customer is near a major airport. For this service, the customer is expected to pay shipping charges.

IMPORTANT: When shipping a module for repair or exchange, always call Lexicon before packaging it for shipment; Lexicon ships modules in reuseable static protective bags with appropriate packing materials -- use these materials or procure new materials from Lexicon. Lexicon is not liable for damage resulting from unauthorized shipping procedures.

2.4 Ordering Parts

Replacement parts can be ordered from:

Lexicon, Inc.
60 Turner Street
Waltham, MA 02154 USA
(617) 891-6790
Telex 923 468
Attn: Customer Service

2.5 Limited Warranty

Lexicon warrants each 224X to be free from defects in material and workmanship under normal use and service for one year. This warranty begins on the date of delivery to the purchaser or his authorized agent or carrier. During the warranty period, Lexicon will repair, or at its option replace, at no charge, components that prove to be defective, provided that the equipment is returned, shipping prepaid, to Lexicon's factory or designated service facility.

The warranty is null and void under the following conditions:

1. Abuse, neglect, alteration, or repair by unauthorized personnel.
2. Damage caused by improper use or operation from an incorrect power source.
3. Damage caused by accident, act of God, war, or civil insurrection.

Lexicon is not be responsible for loss or damage, direct or consequential, resulting from machine failure or the inability of the product to perform. Lexicon is not be responsible for damage or loss during shipment to or from its factory or designated service facility.

Lexicon reserves the right to make changes or improvements in the design or construction of the machine without obligation to make such changes or improvements in the purchaser's machine.

No equipment may be returned under this warranty without prior authorization from Lexicon. Shipments must be packed in authorized Lexicon packing material, fully insured, and prepaid.

This warranty is in lieu of all other warranties, expressed or implied, and of any other liabilities on Lexicon's part; in addition, Lexicon does not assume or authorize anyone to make any warranty or assume any liability not strictly in accordance with the above.

3 THEORY OF OPERATION

3.1 System Overview

As shown in Fig. 3.1, the 224X is divided into 11 major functional modules:

- 1 Control Head
- 2 Transition
- 3 Single-Board Computer (SBC)
- 4 Nonvolatile Storage (NVS)
- 5 Timing and Control (T&C)
- 6 Data Memory (DMEM)
- 7 Arithmetic Unit (ARU)
- 8 Floating Point Converter (FPC)
- 9 Audio Input (AIN)
- 10 Audio Output (AOUT)
- 11 Power Supplies (PS1, PS2, and PS3)

Except for the control head, Transition module, and power supplies, all modules plug into an 8-slot card cage and are interconnected via a motherboard. The card cage and the power supply are contained in a mainframe. The control head is connected to the mainframe by a 25-conductor cable. The Timing and Control (T&C), Data Memory (DMEM), and Arithmetic Unit (ARU) modules comprise a dedicated, 293-ns-cycle, microprogrammed digital signal processor (DSP).

During normal operation, signal flow begins with the two audio input channels. The Audio Input (AIN) module filters, samples, and digitizes analog audio signals into 14-bit floating point representations (12-bit mantissa, 2-bit exponent). These floating point representations are then converted into 16-bit fixed-point two's complement numbers by the Floating Point Converter (FPC) module. The DSP processes this information and generates up to four independent channels of output data. This processed data is passed back to the FPC module, which reconverts it into floating point form. To generate the audio output, the Audio Output (AOUT) module reconstitutes four analog signals from the digital data stream.

The Single Board Computer (SBC) module is a controller that interfaces the control head to the DSP. The microprocessor (an 8080) on the SBC module scans the switches and pots on the control head and drives the control head displays. It processes the information received from the control head and changes the program running in the DSP. In addition, the SBC module performs various housekeeping tasks, such as power-up diagnostics and storing and retrieving nonvolatile user setups from the Nonvolatile Storage (NVS) module.

3.2 Control Head and Transition Module

The control head contains the switches and slide pots that allow a user to modify the control parameters of the reverberation and effects programs and control program access. In addition, the control head displays program and input signal level information.

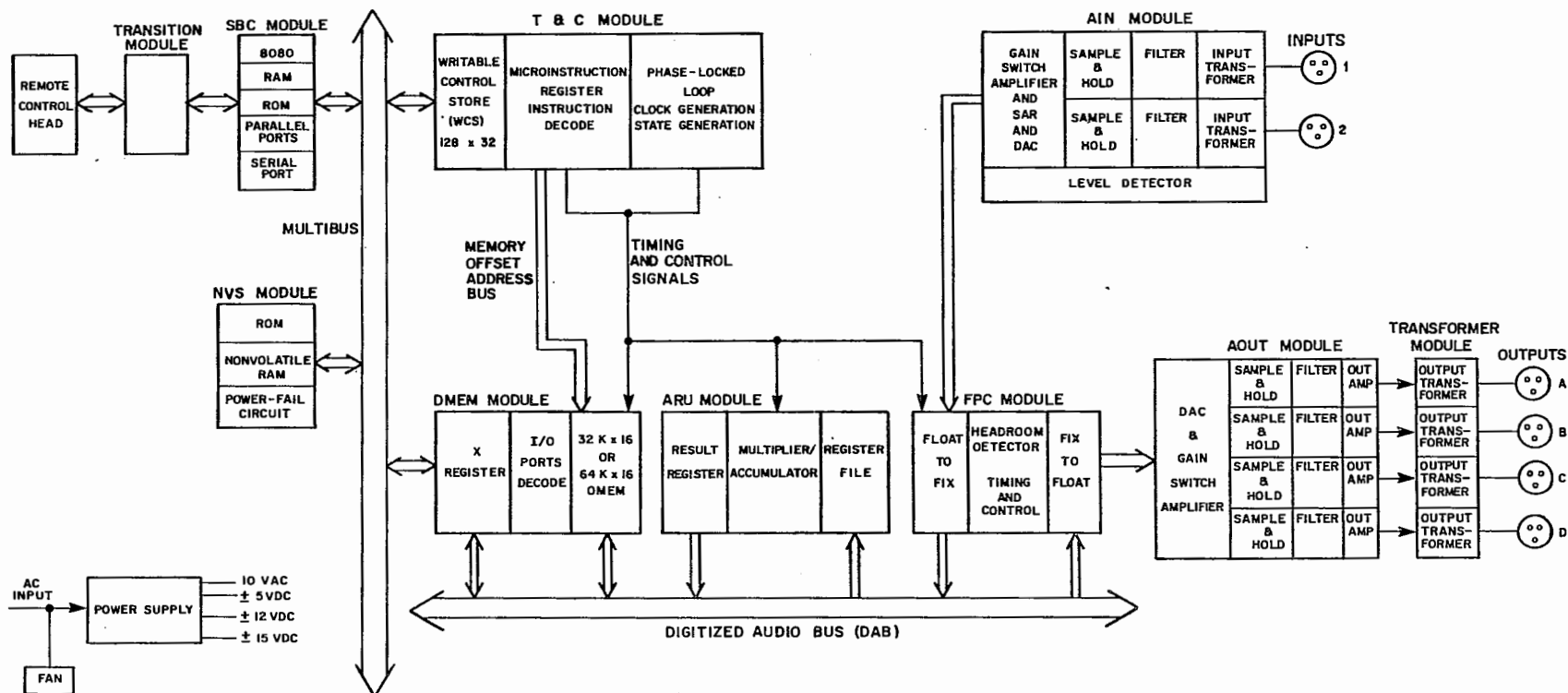


Fig. 3.1. Detailed Block Diagram -- 224X.

The control head consists of two board assemblies connected by a 27-pin flexible cable. The display and pushbutton board has three diodes, eight current-limiting resistors, switches, and LED displays. The second board contains all the remaining electronics, including the slidepots, and a 25-pin I/O connector. The I/O connector has 14 signal wires and two power wires. The rest of the 25 lines are used for digital and chassis ground returns. Ground returns are inserted between signal lines to prevent signal interference. The 14 signal wires connecting the control head to the mainframe consist of three groups of signals from three I/O ports on the SBC module: eight lines from port A form a bidirectional data bus; four lines from port B form a 4-bit address; two lines from port C serve as control signals. Table 3.1 lists the cable and connector wiring runs and tie points between the control head and the SBC module.

Table 3.1. Control Head to Mainframe Wiring.

25-pin Connector (Control Head to Mainframe)	50-pin Connector (Transition to SBC Module)	Function Name
1	2	GND
2	1	PB3
3	3	PB2
4	5	PB1
5	7	PB0
6	4, 6, 8	GND
7	*	GND (Chassis)
8	20	GND
9	21, 29	PC1
10	30	GND
11	25	PC0
12	26	GND
13	33	PA7
14	35	PA6
15	37	PA5
16	39	PA4
17	47	PA3
18	45	PA2
19	41	PA1
20	43	PA0
21	34, 36, 38, 40, 42, 44, 46, 48	GND
22	**	AC1
23	**	AC1
24	**	AC2
25	**	AC2

* Connected to chassis, not to SBC module.

** Connected to 10-Vac secondary, not to SBC module.

The control head interfaces with the SBC module through a 25-pin cable, with a transition at the mainframe to a 50-pin cable: this cable, in turn, connects to the J1 edge connector on the SBC module. The transition from 25 to 50 conductors is made through the Transition module, which is also used as an input point for the AC power to the control head, or 10-Vac power supply generated from a separate secondary of the power transformer. The 10-Vac power supply is fused to protect against cable shorts or similar faulty conditions.

Power Supply. The power supply for the control head consists of a full-wave bridge with filter and a 5-V 7805 regulator. The unregulated dc supply directly powers the LED displays.

Rectification at the control head allows power transmission to the control head without IR voltage drop in the ground returns, because a voltage drop in the ground returns would degrade noise margins and increase noise spikes in the logic signals.

Display Section. The display section is based on a scanned display concept in which all digits share a common segment drive. All digits use a common anode, and current flows through digit segments only if the cathode for that digit is grounded. The display and data transfer sections of the control head use as few interconnection wires as possible. The segment data comes out on port A; the 4-bit address selecting the digit comes out on port B; the control signals come out on port C.

The display cycle is controlled by software. At the end of a cycle, the eight lines of port A are used to send pot or switch data back to the SBC module. Addresses for the switches and pots are the same as for the respective digit just used during a display cycle. Table 3.2 lists the various display and read-back device addresses. The command sequences used are as follows:

ACTION

```
Load Digit, Port B
Load Segments, Port A
Set PC2/ (ACK/)
Set PC0 (STCONV)
Clear PC0
.
Delay Approximately 500 us
.
Clear PC2/
Set PC1/ (Read Enable)
.
Delay 12 us
Read Data
Clear PC1/
.
.
Select Next Digit
Repeat Sequence
```

Table 3.2. Control Head Displays, Pots, and Pushbuttons.

Port B 4-Bit Addresses																			
Port A 8-bit data bus		0		1		2		3		4		5		6		7		8	
		0000		0001		0010		0011		0100		0101		0110		0111		1000	
		Write	Read	Write	Read	Write	Read	Write	Read	Write	Read	Write	Read	Write	Read	Write	Read	Write	Read
		Digit 0	Bass-Pot	Digit 1	Mid	Digit 2	Cross-over	LEDs	Treble Decay	LEDs	Depth	LEDs	Pre-Delay	LEDs	Push-button	LEDs	Push-button		Push-button
PA0		SEG a	LSB	SEG a	LSB	SEG a	LSB	PROG 1	LSB	IMEO	LSB	SEC	LSB	L 0dB	PROG 1	R 0dB	IMED	---	BASS
PA1		SEG b		SEG b		SEG b		PROG 2		SET		MS		L 6dB	PROG 2	R 6dB	SET	---	MID
PA2		SEG c		SEG c		SEG c		PROG 3		CALL		Hz		L 12dB	PROG 3	R 12dB	CALL	---	CROSS-OVER
PA3		SEG d		SEG d		SEG d		PROG 4		SHIFT		kHz		L 18dB	PROG 4	R 18dB	SHIFT	---	TREBLE
PA4		SEG e		SEG e		SEG e		PROG 5		REG A		OVFL		L 24dB	PROG 5	R 24dB	REG A	---	DEPTH
PA5		SEG f		SEG f		SEG f		PROG 6		REG B		---		BASS-POT	PROG 6	TREBLE DECAY	REG B	---	PRE-DELAY
PA6		SEG g		SEG g		SEG g		PROG 7		REG C		---		MID POT	PROG 7	DEPTH	REG C	---	---
PA7		DP	MSB	DP	MSB	DP	MSB	PROG 8	MSB	REG D	MSB	---	MSB	CROSS-OVER	PROG 8	PRE-DELAY	REG D	---	---

A 74LS42 4-to-10 line decoder, controlling 75376 high-current drivers, selects digits. The LEDs are arranged into eight digits: 0, 1, and 2 correspond to the three 7-segment displays, and 3 to 9 are groupings of the various discrete LEDs on the panel. Refer to the schematics for details. 75327 driver arrays, current-limited with 150-ohm resistors, provide segment drive. The RDENB/ signal disables segment drive when pot or switch data is to be transmitted over the eight data lines.

Each time a display cycle is begun, a one-shot U2 fires to start the 75326 drivers for a period of several microseconds. If the SBC module becomes hung up in an unresolved operation, the one-shot times out, turning off all displays to prevent sustained high current (50 mA) from being drawn to any LED. The LEDs can run at high currents for only brief periods.

Slidepot Digitization. The control head uses an ADC-0817, which is a complete A/D subsystem capable of scanning up to 16 inputs and converting each input amplitude to an 8-bit binary code representation. A clock source, a start conversion pulse, and addresses are the only inputs. The ADC-0817 is used in a ratiometric configuration -- that is, the pot terminals and chip +REF (pin 19) are tied to 5 V, and the low terminals and ground (0 V) are tied to -REF (pin 23). The pots represent the complete range from -REF to +REF as a linear function of the position of the slidepots. The ADC translates this to 0000 0000 - 1111 1111 codes.

The start conversion pulse (STCONV) is filtered for noise rejection and routed to the start-convert and address-latch inputs of the ADC. The ADC synchronizes this command to its free-running input clock and begins a conversion cycle. Within 64 clock cycles, or 128 μ s, assuming a 500-kHz clock, the ADC-0817 completes the conversion and outputs the result at tristate outputs. An internal analog multiplexer selects the desired pot.

The clock to the ADC-0817 A/D converter is generated by a CMOS/RC free-running square-wave oscillator running at a nominal 500 kHz $\pm$ 150 kHz ($\pm$ 30%). The actual conversion time ranges from 70 to 130 μ s. When RDENB/ is LOW and the digit address is 0 through 5, data from the ADC-0817 outputs is gated onto the PA0-7 data bus. A tristate LS244 is used for this function.

Switch Data. The switches are all normally open pushbuttons arranged in three banks corresponding to digits 6 to 8. Germanium diodes 1N283 are used to isolate each bank or column of switches if more than one pushbutton is pressed at the same time.

The banks are wired into rows assigned to data lines PA0-7. Pull-up resistors ensure proper threshold for the tristate buffers. A complete bank or column of switches can be read when RDENB/ is LOW and the bank's digit line is selected. Any pushbutton pressed corresponds to data out onto the bus as a TTL logic "0". Refer to the schematics or Table 3.2 for details of pushbutton and pot assignments.

3.3 Single-Board Computer (SBC) Module

The SBC module is a National Semiconductor BLC-11 (or equivalent) using an 8080 microprocessor. It also includes 1K bytes of RAM at hexadecimal addresses 3C00 to 3FFF and supports four 5-V 2716 ROMs to provide a total of 8K bytes of ROM at hexadecimal addresses 0 to 1FFF.

The SBC module controls all functions of the control head, such as reading switches and slide pots, as well as data display. Its ROMs also contain the reverberation and effects software, which controls the DSP. The multibus provides the pathway for interaction between the SBC and the DSP. The software can be updated and expanded simply by replacing ROMs. The SBC module has three parallel ports and one serial port. The serial port is normally not used, but it has been configured as an RS-232 data set at 4800 baud (settable from 110 to 9600 baud).

Parallel port A is a bidirectional port and is used in mode 2. Parallel port B is used in mode 0 and provides four output bits used as an address to the control head. Parallel port C provides both control bits for port A and the control head.

3.4 Nonvolatile Storage (NVS) Module

The NVS module (sometimes referred to as the memory expansion module) provides nonvolatile memory consisting of (1) battery-backed-up RAM for the 36 user storage registers that hold customized reverberation and effects program variations, and (2) ROM storage expansion for the reverberation and effects software. The NVS module is contained on a single multibus-compatible board that plugs into the OPTION slot on the 224X card cage.

A NiCad battery backup powers the RAM of the NVS module to preserve the memory contents after power has been shut off. The RAM consists of four 1K x 4-bit RAMs (U5, U6, U12, U13) for a total of 2K bytes. It resides in memory address from hexadecimal 2000 to 27FF. The battery backup can protect the memory contents for as long as three months without recharging. All control head settings and register stores can be saved. The batteries are maintained at full charge by a trickle charger that operates when the 224X is turned on. The charger consists of a 78L05 that regulates the +12-V supply down to +5 V plus a diode drop. This voltage powers the battery supply Vbb via CR1 when power is on. The battery is charged via R16, a 270-ohm resistor. The batteries are fully charged after power has been on for 20 hours. An on-board detection circuit monitors the ac mains power and places the memory in protected store mode when a power outage is detected.

The 10-Vac power supply is conditioned to a TTL-compatible level to trigger the one-shot U21. Under normal power-up, the one-shots are continuously triggered, thus disabling the power-fail signal PFAIL. When a power failure occurs, the one-shot is not triggered and PFAIL will be activated. When PFAIL is activated, the write signals to the nonvolatile RAMs are disabled, preventing any accidental erasure to the contents. Part of the power fail circuitry is powered from the battery voltage Vbb to prevent any unpredictable behavior at power fail.

The ROM section of the NVS module has sockets for eight 2732-type 4K x 8 ROMs for storage of reverberation and effects software. The ROM resides in memory address from hexadecimal 8000 to FFFE. Memory location FFFF is mapped to a 4-position dip switch. (Thus the last byte in the last ROM cannot be read by the microprocessor.)

The dip switch may be preset to one of 16 code combinations, corresponding to 15 registers. The software reads the switch setting upon power-up to determine which register, if any, has been selected. By changing the switch settings, the 224X can power up with a predetermined program setup.

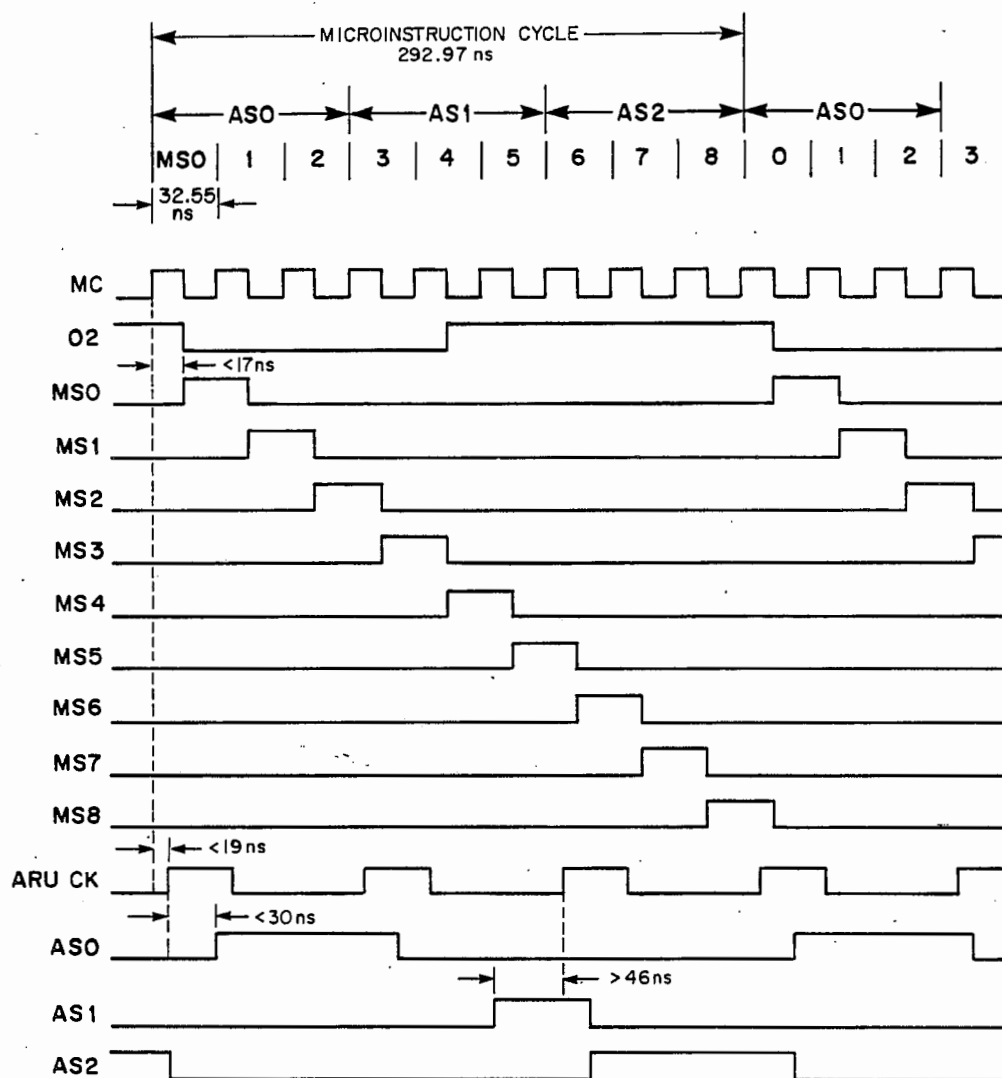
3.5 Timing and Control (T&C) Module

The T&C module contains the clock and state generation circuitry, 512 bytes of microprogram memory, an interface between the SBC module and the microprogram memory, and microinstruction decode and control signal generation logic.

The clock generation circuitry on the T&C module is synchronized to the SBC module by a phase lock loop. ϕ_{27} is the clock signal generated by the SBC module, running at 2.048 MHz. An MC4044, U27, forms the phase comparator and the low pass filter. The voltage controlled oscillator is formed by an LC tank oscillator, with a varactor, CR1, for voltage control. A divide-by-15 counter, U41, is embedded in the phase lock loop, making it a frequency multiplier of 15. Thus the master clock, MC, runs at 30.72 MHz. Since the LC tank oscillator output is divided by 2 by a J-K flip-flop, U26, to become MC, the oscillator actually runs at 61.44 MHz. This clock can be observed at TP2 (OSC). MC is driven from a Schottky nand gate, U40, with an active pull-up, Q3.

The state generation circuitry consists of a divide-by-9 counter, U56, and an 8-bit shift register U39. Thus the system clock period is at 3.41 MHz, i.e., 293 ns, which is divided into nine time slots, MS0-MS8. These nine time slots are grouped into three, AS0, AS1, and AS2 by U38, U25, and U23. Refer to Fig. 3.2 for the basic timing of the T&C module.

The control signal generation is achieved through a writable control store (WCS), formed by four 128 x 8 static RAMs (U2, U15, U29, U43). The WCS is cycled by an eight bit program counter (U1, U14). A 100-step control program is allowed, giving a sampling rate of 34.13 kHz. Thus the counter is normally reset at count 99 by a RESET signal that is generated by the WCS itself. The WCS program is loaded from the SBC module. Thus the address from the counter is multiplexed with the address from the SBC module. The bidirectional data bus to the WCS is buffered by bidirectional drivers, U3, U16, U30, U44, to the SBC module.



NOTES:

ARU OPERATES ON A CLOCK THAT CAN LAG MC BY $< 19 \text{ ns}$.
 AS1 IS SHORTER THAN EITHER AS0 OR AS2 TO AVOID
 MS4 CLEARING AS0 PREMATURELY.

Fig. 3.2. T&C Module Timing.

The least significant 16 bits of the WCS word are normally used as an address for the data memory. The remaining 16 bits contain the multiplier coefficient, the register file addresses, and miscellaneous control signals. They are clocked by a 32-bit microinstruction register, U4, U5, U17, U18, U31, U45. The upper 16 bits of the microinstruction register are formed by 74S163 counters for the synchronous clear functions that they provide. The upper 16 bits of the microinstruction register are cleared whenever the cycle is used by the SBC module in accessing the WCS, since the data on the microinstruction bus, MIO-MI31, may not be valid. The microinstruction word is set up such that all zeros correspond to a no-operation. The lower 16 bits of the microinstruction word can be "don't cares," since they represent only an address.

The remaining control signal generation circuitry consists of decoders (U47, U48, U49, U32, U34), which generate the control signals that are encoded in the instruction word; register U19, U20, which further pipelines and synchronizes the required control signals; and flip-flops U24, U22, U21, U20, which generate more complicated control signals from the basic timings MS0-MS8. The multiplier coefficient from the microinstruction word, C0/ to C5/, is serialized into an even and an odd stream, M0/ and M1/, by shift registers U10 and U11 for the serial multiplier in the ARU module.

The access to the WCS from the SBC module is decoded by U50 and synchronized by U52 and U53. Since the multibus is an asynchronous bus, an acknowledge signal to the SBC module, XACK/, is generated by U54. The WCS can be accessed from the SBC module in two modes. First, the DSP can be halted by stopping the program counter (U1, U14) by asserting HALT/. In this mode, the SBC module can read from and write to the WCS anytime. The WCS is mapped into hexadecimal address locations 4000 - 41FF of the SBC module address space. The decoder U46 decodes the least significant two bits of the address from the SBC module to select one out of the four 128 x 8 bit static RAMs and the corresponding bidirectional drivers. Alternatively, the SBC module can also access the WCS while the DSP is running, allowing it to change program characteristics on the fly. In this mode, a protect bit in the microinstruction determines when the SBC module can access the WCS. The program must be organized such that the microinstruction, which is displaced during an SBC access, is a no-operation. Normally, the SBC module only writes into the WCS in this mode, which only takes up one microinstruction time. Reading from the WCS in this mode can take up to three microinstruction times. In this mode, a flip-flop, half of U53, is used to attempt to synchronize pairs of SBC module accesses in the same sample time: an allowed access slot not utilized will disable the next access slot.

Finally, there is some diagnostic hardware included in the T&C module. Three groups of eight timing control signals can be read by the SBC module via tristate bus drivers, U6, or registers U7 and U8. The DSP is halted when these diagnostic ports are being read. In the registers U7 and U8, the signals being read are dynamic and thus need to be sampled by appropriate clocks. Note that the digital overload signal SAT is also read through one of these diagnostic ports, U8. This allows the SBC module to detect when digital overflow has occurred in the ARU module. The SAT signal is first clocked through a flip-flop. Then it triggers a one-shot since the SAT signal can be transient. When the overflow is continuous, the one-shot is not triggered. Thus the SAT signal is OR-ed with the output of the one-shot, which is then read by the SBC module. U9 forms a shift register to perform a serial-to-parallel conversion of the four serial bit streams S0, S1, M0/, M1/ to be observed. Exclusive or gates U12 minimize the hardware needed to observe the bit streams. ARUCK fires one-shot U13, the output of which drives an LED on the edge of the board. This gives a preliminary indication of whether the clock is running. Another test point TP1 is available for test synchronization purposes.

3.6 Data Memory (DMEM) Module

The DMEM module contains the data memory, control signal and address generation circuitry, the XREG (DMEM transfer register), diagnostic ports, and the 8080 port-decoding circuitry. The DMEM communicates with the rest of the system over the digitized audio bus (DAB).

The address to the data memory coming from the microinstruction is in the form of an offset relative to a current position in memory. This current position is held by a 16-bit current position counter (U51 and U65) and is normally incremented once every sampling interval. The absolute address of a memory reference is computed by subtracting the offset from the current position. A 2's complement subtraction is performed by adding the complemented 16-bit word, OFST/, to the output of the current position counter and tying the carry input of the adder (U49, U50, U63, and U64) high. A multiplexer (U18 and U36) is used to multiplex the resulting address onto the eight address lines of the 64k dynamic RAMs. The circuitry is set up such that either one bank of 64k dynamic RAMs or two banks of 16k dynamic RAMs can be used. The address and control lines of all the RAMs are tied together. Because the RAM outputs are capable of fanning out to ten low-power Schottky TTL loads, they are tied directly to the DAB without buffering.

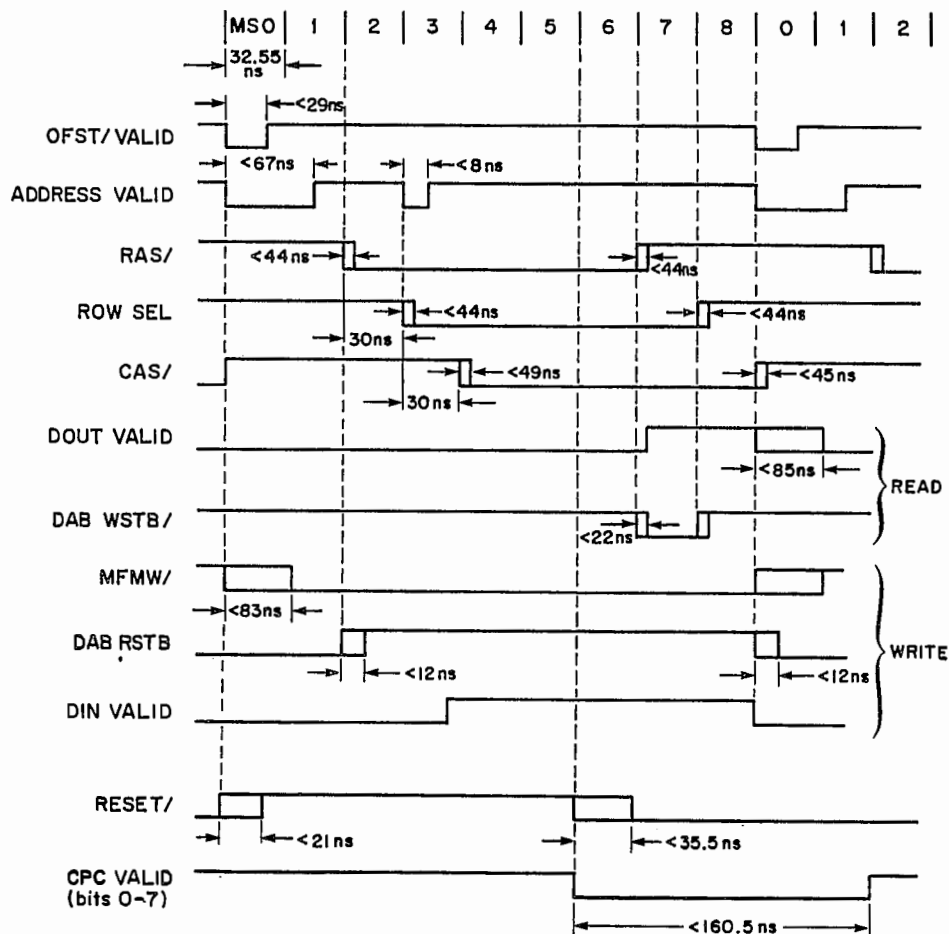
The timing and control signals for the DMEM modules are generated by a delay-line circuit (U59) based on signals supplied by the T&C module MEMAC, DABSTB, and MEMR. Refer to Figure 3.3 for the data memory timing.

In addition to the data memory circuitry, the DMEM module also contains some decoders (U55, U56, U57) which are used to generate the strobes used in the I/O access of the ports used in various modules in the DSP from the SBC module. The open collector-gate U52 is used to return an acknowledge, XACK/, to the SBC module after an I/O access. The NAND gates U53 and U54 are used to implement the single cycle/halt/run control modes of the DSP.

The module can single cycle, halt, or let continue run the DSP through accessing these latches via the I/O ports.

Some diagnostic circuitry is also included on the DMEM module. The tristate bus drivers U48 and U62 are used to enable the SBC module to read the OFST/ lines when they are static. U42 forms the bus test register, which enables the SBC module to sample and read its own data bus DATA/ on the DMEM module. U38, U39, U40, and U41 form the X register, which enables the SBC module to read from and write to the DAB.

U38 and U40 are used to send data from the DAB to the SBC module and U39 and U41 are used to send data from the SBC module to the DAB.



NOTES:

CAS/ FALLS ONLY WHEN MEMAC IS HIGH, INDICATING MEMORY OPERATION.
 CRITICAL TIMING PATH FOR DIN IS XFER CK TO RESULT REGISTER OF ARU.

Fig. 3.3. DMEM Timing.

3.7 Arithmetic Unit (ARU) Module

The ARU consists of a 4 x 16-bit register file, a 16 x 6-bit 2's complement multiplier with saturation logic, a 20-bit accumulator, and a 16-bit result register. The 4 x 16-bit register file acts as a temporary store for the multiplicands taken from the DAB. The source of the multiplicand can thus be from the FPC module, DMEM module, the SBC module via the X registers on the DMEM module or even from the result register. The multiplier performs a 16 x 6-bit multiply and accumulate every system clock time (i.e., 293 ns). The 6-bit multiplier coefficient and the control signals to the multiplier are generated from the T&C module. The result register acts as a buffer between the outputs of the multiplier and the DAB, allowing the multiplier to perform the next multiplication without having to wait for its previous result to be read by the other parties on the DAB. In a similar manner, other parts on the ARU are pipelined to maximize the operating speed of the essentially serial multiplier through the register file, the partial product register, and the accumulator.

The 4 x 16-bit register file (U29, U30, U31, U32) has independent write addresses (WAO, WA1) and read addresses (RA0, RA1), which are controlled by the microinstruction. In this way, data on the DAB can be written into the register file at one address while data at another address can be read by the multiplier. Note that the write signal to the register, DAB WSTB/, is active every system clock time (293 ns) although the data on the DAB is not always relevant. Address 3 in the register file is used as a pass-through location in these instances.

The multiplier is implemented by a modified shift and add serial multiply technique. Instead of the normal shift and add, two shifts and adds are performed at the same time such that the multiply is twice as fast. A system cycle time (293 ns) is divided into three ARU states: AS0, AS1, and AS2. During each of these states, a double shift and add is performed. This gives a 6-bit multiply in a system cycle time. The double shift is performed by a "dual rank" shift register (one which shifts by two bits at a time) (U3, U4, U15, U16, U17, and U18). The double shift is performed by interleaving the bits to two sets of the shift register: U4, U18, and U15 form one shift register that receives the even-numbered bits, and U3, U17, and U16 form another register that receives the odd-numbered bits. In the first ARU state, AS0, the contents of the register file are simply loaded into the shift register. The output of the shift register is split into two groups: one is the direct output and the other the output shifted right by one bit. Depending on the 6-bit multiplier coefficient from the microinstruction, which is serialized into bit streams M0/ and M1/ in the T&C module, these two are blanked or added to the other (by the nand gates U14, U26, U27, U28, U40, U41, U50, U51, U52, and U53). The result is loaded into the partial product register (U10, U11, U12). Note that the adder forming the partial products is used as a negative logic adder because the nand gates provide an inversion, and the carry input of the adder is tied high. This inversion is taken into account by the following stage of exclusive or gates.

The partial product register acts as a pipeline register for the second stage of addition. As the double shift and add circuit previously described proceeds to perform another double shift and add in the second ARU state, AS1, the partial product is added to the accumulator. The outputs of the partial product register are passed through a set of exclusive or gates (U5, U6, U7, U8, and U9) controlled by the sign bit of the multiply CSIGN/. Depending on the logic state of CSIGN/, the data can be negated by inverting the data and tying high the carry input of the adder that follows. U19, U20, U21, U22, and U23 form the adder that adds the partial product to the accumulator.

Overflow in the ARU has to be handled properly. In this system, saturation arithmetic is performed. In the event of a positive or negative overflow, the most positive number or the most negative number is forced in place of the overflow number. This is implemented by the most significant two bits in the 20-bit data path within the multiplier. It should be noted that, in forming the 20-bit word to the multiplier from the 16-bit data from the register file, the most significant bit of the 16-bit word is the two most significant bits in the 20-bit word and the

least significant three bits of the 20-bit word are tied to zero. Thus, the most significant two bits of the 20-bit word should always be the same unless an overflow has occurred. This condition is detected by exclusive or gate U42. This would force the multiplexors (U33, U34, U35, U36, and U37) to select either the most positive or the most negative numbers, depending on the MSB of the overflow number. The accumulator is formed by 4-bit counters 74LS163s (U45, U46, U47, U48, and U49). The counting function is not used, however; only the synchronous load and clear functions are used.

Because of pipelining, the final result of the multiply and accumulate does not become available until the very end of ASO of the next system cycle. If a transfer to the result register command is present in the microinstruction, the result register is loaded at this time by XFER CK. If the zero accumulator command is given, the accumulator is also cleared at this instant. Refer to Fig. 3.4 for the ARU module timing.

3.8 Floating Point Converter (FPC) Module

The FPC module serves as an interface between the DSP and the two analog modules: AIN and AOUT. For the DSP, the analog I/O looks simply like another device that can be read from and written into via the DAB. For the analog modules, the FPC module is the source of the timing strobes and clocks that direct the A/D and D/A conversion processes. In addition, the FPC module is responsible for making the floating point/fixed-point translation required for communication between the two analog modules and the DAB.

The FPC module has four major functions:

1. Timing and control generation
2. Input floating-to-fixed-point conversion
3. Output fixed-to-floating-point conversion
4. Headroom level indication.

Timing and Control Generation Circuitry. Besides supplying its own timing and control signals, the FPC module generates the timing and control signals for the analog modules. It generates these signals based on several signals from the T&C module: FPC CK, which occurs every system clock time (293 ns), provides the basic clock; RESET/, which occurs every sample time (29.3 ns), synchronizes the timing cycles; RD AD/ and WR DA/, which control the reading from and writing to the floating-to-fixed-point circuitry. The timing and control signals are generated by an 8-bit input cycle counter (U7 and U8), which drives a 256- x 4-bit ROM (U6). Some of the outputs of the ROM are deglitched by a register (U18), and some are directly used as timing and control signals. Because 100 system clock times occur in each sample time, only the first 100 locations of the ROM are used.

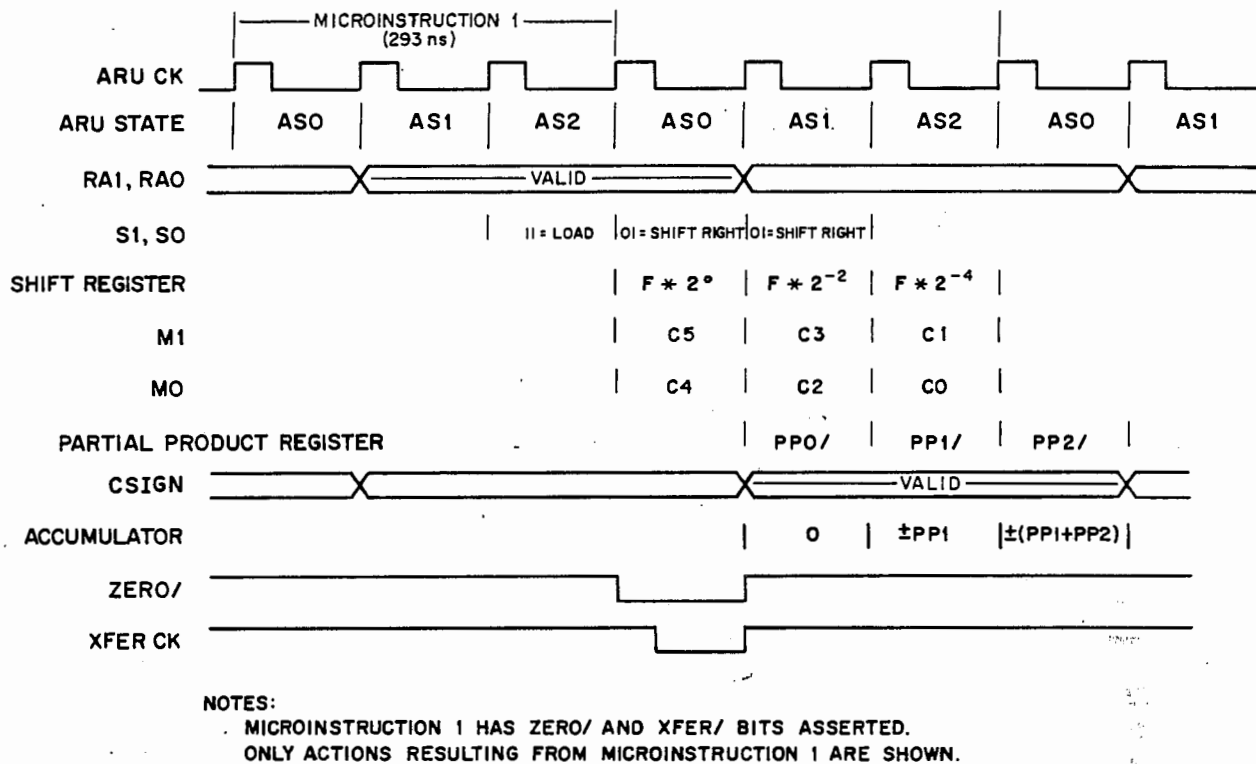


Fig. 3.4. ARU Module Timing.

Two multiplexors (U4 and U42) implement a self-test mode in which the FPC CK signal is replaced by 02/, the SBC system clock (488 ns) and the RESET/, RD AD/, WR DA/, and the output channel select signals SDAA, SDAB< SDAC< SDAD, are substituted by signals decoded from the counter by nand gates (U5).

The select signal to the multiplexors, FPC DEBUG, is tied low in the T&C, ARU, and DMEM modules. Thus, when these modules are removed, the FPC DEBUG signal floats high and the self-test mode is automatically activated. In the self-test mode, the only external signal needed is the 02/ clock. The input from the left channel is immediately transmitted to the output channels A and B via the FPC module. Similarly, the right channel is transmitted to channels C and D. In this way, the FPC, the AIN, and the AOUT modules can be checked independently of the DSP. However, in using the 02/ signal from the SBC module, the sampling frequency changes from 34.13 kHz to 20.48 kHz. To obtain the correct sampling rate, the SBC module must be removed also and a 3.413-MHz TTL-compatible input clock must be supplied to pin A28 of the backplane connector using a signal generator.

Input Floating-to-Fixed Point Conversion Circuitry. At the beginning of a sample cycle (starting with state 0 of the input cycle counter), the successive approximation register (SAR) on the AIN module is instructed to start an input conversion by bringing STCONV/ signal high and sending thirteen clock pulses on the CNVCK signal. Just after the twelfth clock pulse, the SAR contains valid input data. The LOAD signal to the shift registers (U27, U28, U38, and U39) is then brought high, loading the input data into the shift registers. STCONV/ is brought low again and the thirteenth clock pulse on the CNVCK resets the SAR, readying it for the next conversion.

Shift register U27, U28, U38, and U39 and counter U16 perform the floating-point-to-fixed-point conversion. Note that the LOAD signal is asserted for two clock pulses. The first clock pulse loads the input gain counter, U16, with the input gain bits IGA1 and IGA0. Because a one is also loaded into bit 2 of the input gain counter (U16 pin 12), which is connected to the S1 input of the shift register, both S1 and S0 of the shift register are high for the second clock pulse, causing it to load the 12 bits from the SAR. When LOAD returns low, the input gain counter counts up, and the shift register shifts left until QC QB QA = 000. Gain bits of 00 result in four shifts; 01 results in three shifts; 10 in two shifts; and 11 in one shift. By the fourth clock pulse following the falling edge of LOAD, the Channel 2 conversion is complete. Tristate drivers U25 and U26 enable this data onto the DAB when the DSP asserts the RD AD/ line. Meanwhile, CH1L (a signal derived from CH1 on the AIN module) goes high, causing an input channel switch. A similar conversion for Channel 1 takes place during the second half of the input conversion cycle. Refer to Fig. 3.5 for the floating-to-fixed-point conversion and A/D conversion cycle timing.

Output Fixed-to-Floating-Point Conversion Circuitry. The 4-bit register, U40, and the 16-bit register, U36, U37, comprise a double buffer that stores the output channel select code, SDAA-SDAD, and the 16-bit fixed point output value from the DSP, respectively. When the DSP signals an output to the D/A by asserting WR DA/, the double buffer is loaded and NEW DAT/ (U3 pin 6) is asserted, indicating that the double buffer is full. At the next clock pulse, BUSY from the strobe counter, U1, U2, is inspected. If BUSY is high, indicating that a D/A conversion is currently taking place, nothing happens. If BUSY is low, the 16-bit data stored in the double buffer is loaded into shift register U23, U24, U34, U35; the 4-bit select code stored in the double buffer is loaded into register U41; and the flag NEW DAT/, is deasserted, indicating that the double buffer is ready for the next output value. NOR gate U14 pin 1 ensures that this flag is not deasserted if the DSP reloads the double buffer, just as the old information in the double buffer is loaded into the shift registers.

When the shift register is loaded, strobe counter, U1, U2, and output gain counter U43 are loaded, initiating an output cycle. Refer to Fig. 3.6. The output gain counter counts up from zero and the shift register shifts left until NOR gate U14 pin 4 detects one of two conditions: (1) The sign bit is about to be shifted out of the shift register (that is, the two MSBs disagree). When STOP/ goes low, the fixed-to-floating-point conversion is complete. (2) The counter has incremented three times.

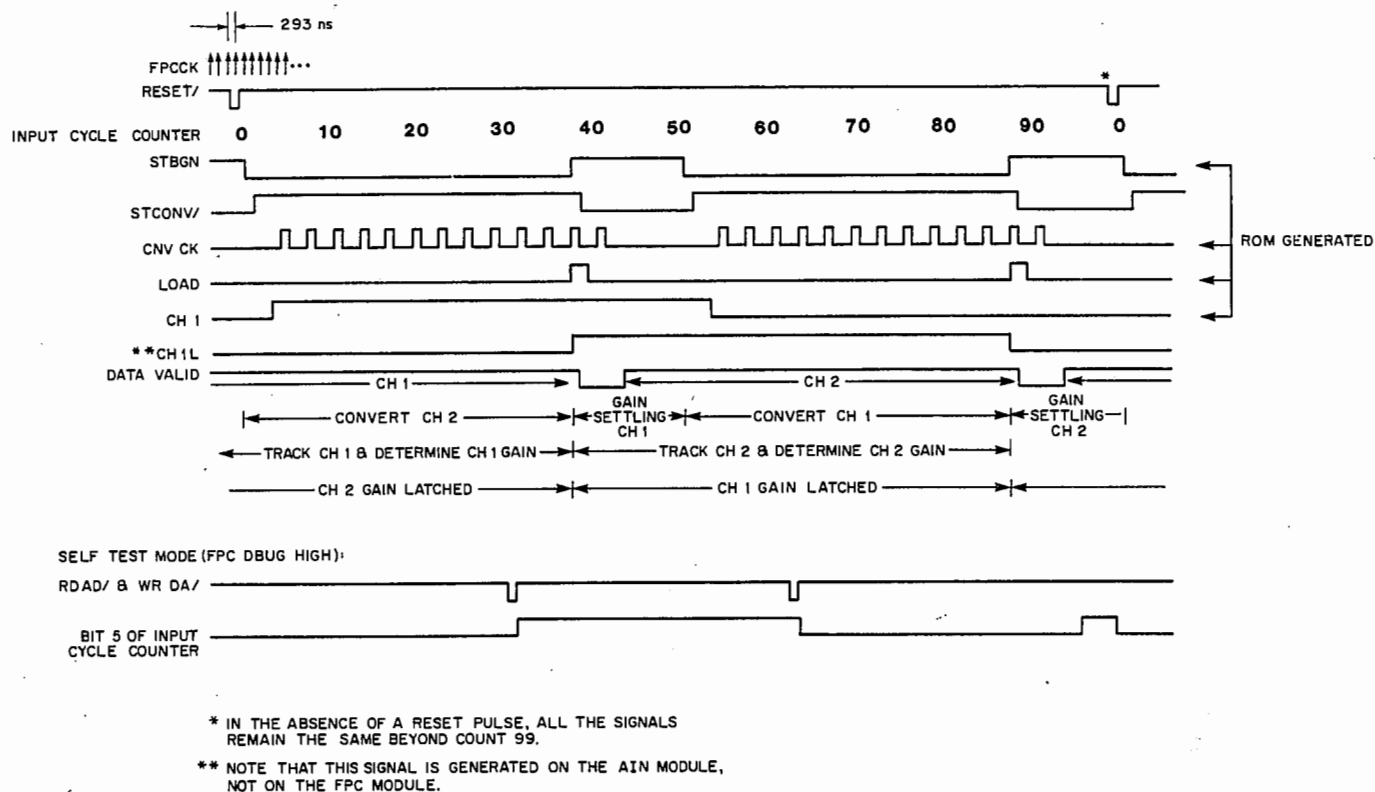


Fig. 3.5. Floating-to-Fixed-Point Conversion and A/D Cycle Timing.

The two LSBs of the output gain counter are transmitted to the gain switch amplifier (GSA) on the AOUT module as output gain bits OGA1, OGA0. Meanwhile, the strobe counter has been counting up from its initially loaded value of hexadecimal 2A. After allowing enough time for the fixed-to-floating-point conversion to complete and then enough time for the GSA and D/A converter on the AOUT module to settle, flip-flop U3 pulses the multiplexor enable U42 pin 15, thereby strobing the appropriate output line OUTA-OUTD.

Headroom Level Circuitry. The headroom level information is sent to the FPC module from the AIN module, where rectifiers and comparators generate a 5-bit code representing the instantaneous levels of the analog inputs. Because this information is multiplexed between the two input channels, the FPC module demultiplexes it by clocking two headroom registers, U31, U32 and U21, U20, on opposite edges of the channel select signal, CH1. Peak detection occurs by clearing a register bit any time the corresponding headroom bit is asserted. What remains in the register, then, is the complement of the largest headroom word that has occurred since the register was initialized. The SBC module reads a headroom register by strobing either HR1/ or HR2 low, enabling the tristate drivers U19 or U30 onto the SBC module data bus. The rising edge of the strobes trigger a one-shot, which initializes the corresponding headroom register.

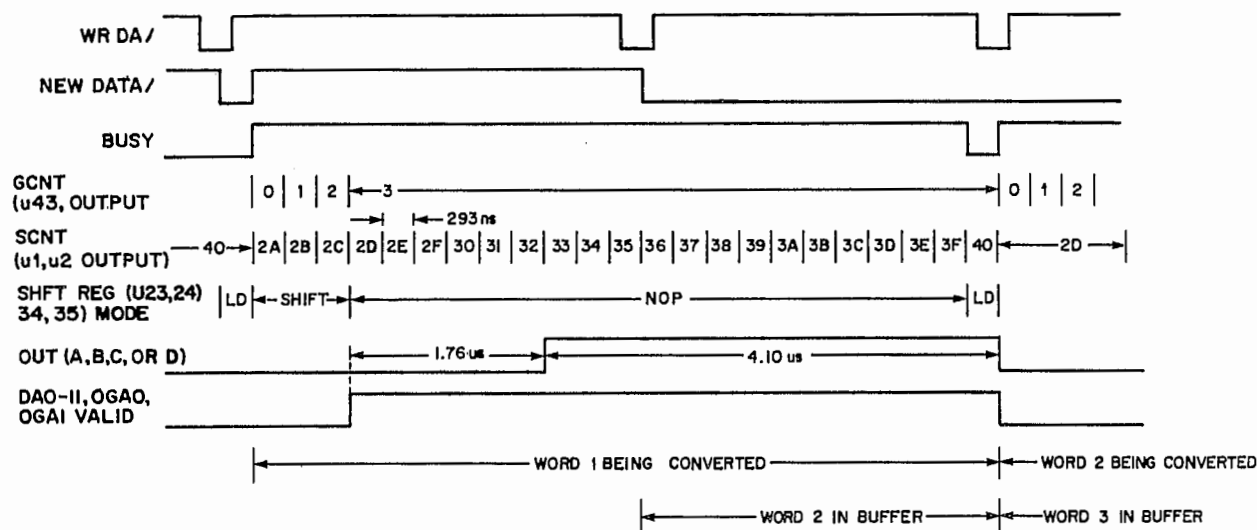


Fig. 3.6. Fixed-to-Floating-Point Conversion and D/A Cycle Timing.

3.9 Audio Input (AIN) Module

The AIN module gain conditions, filters, and digitizes two input channels of audio signals in a floating point format. The major subsections of the module are:

1. Input gain conditioning and filtering
2. Sample-and-hold and multiplexer circuits
3. Gain ranger
4. Analog-to-digital converter (ADC)

Input Gain Conditioning and Filtering. Inputs are transformer-coupled and then gain-conditioned by a buffer stage with an adjustment range of 15 dB. The nominal level at the output of buffer stage U1 is +13 dBm (5-V peak) at 1 kHz. This amplitude corresponds to the onset of clipping in the ADC. Diode clamps prevent overloading the input stage of U1.

The input filters are 7-pole active elliptical (or Cauer) networks synthesized from "FDNR" networks. The nominal cut-off frequency for the input filters is 15 kHz to prevent aliasing distortion at a sampling rate of 34.13 kHz. The next stage is a shelving preemphasis network with 50-us

and 12.5-us time constants. The last filter stage provides aperture correction to compensate for the slight amount of high-frequency loss introduced by the sampling process.

Sample-and-Hold and Multiplexer Circuits. The input sample-and-hold (S&H) circuits are designed so that when one channel is tracking, the other channel is in hold mode. Control signal CH1L places U20 in hold mode when HIGH, simultaneously placing U21 in tracking mode. Analog switch U22 is controlled by HLCH1L, an inverted and level-shifted version of CH1L. This switch is configured so that the "held" channel is commutated to the gain switch amplifier (GSA) stage.

Gain Ranger. Both filtered channels are sent to precision full-wave rectifier circuits that give a positive output equal to the peak amplitude of their inputs. The rectified output of the tracked channel is routed by analog switch U14 to an amplitude quantizer made up of five comparators biased at 5.0 V, 2.24 V, 1.12 V, 0.56 V, and 0.280 V, respectively. These thresholds are arranged so that 5 V corresponds to the onset of clipping in the ADC, and each of the lower thresholds is 6 dB apart and allows determination of the optimum gain to be used for the GSA. Table 3.3 shows how the proper gain is selected for signals of various amplitudes.

Table 3.3. Comparator Outputs Showing Selected Gains for Various Signals.

Signal	U15-13 (5.0 V)	U15-1 (2.2 V)	U15-2 (1.12 V)	U16-1 (560 mV)	U16-2 (280 mV)	GAIN A (dB)	IGA1	IGA0
>5.0 V	0	0	0	0	0	0	0*	0*
2.24- 5.0 V	1	0	0	0	0	0	0	0
1.12- 2.24 V	1	1	0	0	0	6	0	1
560 mV- 1.12 V	1	1	1	0	0	12	1	0
280 mV- 560 mV	1	1	1	1	0	18	1	1
<280 mV	1	1	1	1	1	18	1	1

*Comparator outputs are decoded and latched to provide gain-control signals IGA0 and IGA1.

The 5-V and the 280-mV comparator outputs are needed for headroom display only, while all of the additional comparators provide gain-change information as well as headroom display. A gain of 6 dB is added whenever a signal falls below 45% of the clipping level. A signal level falling to 45% of the clipping level would then be increased by a factor of two to

about 90% of clipping level to take full advantage of the ADC's usable dynamic range. The technique of gain ranging used here is commonly referred to as "instantaneous floating point conversion" -- that is, an appropriate gain is selected just before converting each sample. The gain thus follows the envelope of the converted signal.

Excellent signal reconstruction can be achieved if the ADC is properly offset to deliver a zero code for 0-V input. Good gain matching is provided by using a precision resistor network to set the gain of U23. The resistor feedback ratio is selected by analog switch U24. Because U24 conducts negligible current, errors resulting from switch resistance are also negligible. U24 also provides a decode function for gain bits HLG0 and HLG1. Voltage offset in this stage is not important, provided that it is proportional to gain. Maximum offset due to accumulated errors should never exceed 80 mV at this output stage.

Analog-To-Digital Converter. The analog-to-digital converter (ADC) is a 12-bit successive-approximation type. A 12-bit current output digital-to-analog converter (DAC) is sequenced by a successive approximation register (SAR) chip, U26. This device receives its start command (STCONV) and conversion clock (CNVCLK) from the FPC module. An LM211 comparator is used to compare the DAC's output with the input signal. The output of this comparator is used as the input to the SAR.

3.10 Audio Output (AOUT) Module

The AOUT module provides four output channels serviced from a single timeshared DAC and gain switch amplifier. The AOUT module has four major subsections:

1. Digital-to-analog converter (DAC)
2. Gain ranger
3. Demultiplexor and sample-and-hold circuits
4. Filtering and output gain conditioning

Digital-to-Analog Converter. The DAC, U2, is updated by 12-bit word DAO to DA11. The DAC responds with an analog output in the range of -5 to +5 V.

Gain Ranger. Gain control is set by two bits, OGA1 and OGA0, which control analog switch U3. U3 selects one of four taps from precision divider network RD2. Gain selection is either 1, 1/2, 1/4, or 1/8 $\pm 0.05\%$. U4 operates as a high input impedance follower to prevent loading of the attenuator network.

Demultiplier and Sample-and-Hold Circuits. The corresponding channel-selection signal OUT A, B, C or D goes high approximately 1.76 μ s after DAC and gain data become valid. This delay allows time for the DAC and GSA to stabilize before placing the selected output sample-and-hold in the sample mode. The OUT (sample) command is valid for approximately 4.08 μ s. During this time, U6 switches the output gain-conditioned DAC voltage to capacitor C13, 16, 17, or 20. U7 and U8 provide high impedance buffering to prevent discharge of the sample capacitors during the hold period.

The +7- and -7-V CMOS switch bias is provided by zener diode regulators, CR5 and CR6. Power-on muting is provided by transistor Q9, which is held off for several seconds after power is applied. R159 charges a 22-uF capacitor from -7 to +7 V. Saturation of Q9 places a ground signal on pin 6 of U3, enabling U3 to pass signal.

Filtering and Output Gain Conditioning. Output deemphasis is provided by stages U13 and U21. Filtering is provided by 7-pole elliptical (Cauer) filters. Adjustments to compensate component sensitivity are provided for each section. These settings should never need adjustment if related components are not changed.

The output stage consists of a level adjustment potentiometer and an opamp stage with a complementary output transistor buffer for each channel. Overall, negative feedback is provided around the stage to maintain fixed gain and provide low distortion. A loading network is also provided to maintain high-frequency stability. Each output stage drives an output transformer on the audio transformer board at the rear of the chassis with the audio I/O connectors. The output transformer provides a voltage gain of about 2.5.

3.11 Power Supplies

The three 224X power supplies produce six regulated dc voltages and one unregulated ac voltage.

+5-V and -5-V Power Supplies. The +5-V and -5-V power supplies are derived from a single secondary winding fused in both legs by a pair of 15-amp slow-blow fuses. The +5-V power supply consists of a uA723 regulator, a current boost transistor, and a pair of high-current pass transistors. The regulator is a current foldback design that limits short-circuit currents to less than 3 A.

Over-voltage protection is provided by a crowbar circuit. The +5-V power supply is designed to provide a continuous 10 A. Both voltage and current limits are adjustable.

The -5-V power supply is a 7905 monolithic regulator fused at 2.5 A. This power supply is both current-limited and thermally protected. It is designed to provide 250 mA.

+12-V and -12-V Power Supplies. The +12-V and -12-V power supplies are derived from a single secondary fused in both legs by a pair of 3-A slow-blow fuses. The +12-V power supply consists of a LM317K monolithic voltage regulator controlled by a pair of 1% resistors. This power supply is both thermal- and current-protected, and provides 1.25 A. The -12-V power supply is derived from a 7912 monolithic voltage regulator that is both thermal- and current-protected, and provides 150 mA.

The +12-V power supply is interlocked with the -5-V power supply so that +12 V is not available until after -5 V is available. Should any problems occur with the +12-V power supply, check to be sure that the -5-V power supply is available.

+15-V and -15-V Power Supplies. The +15-V and -15-V power supplies use a tracking design that allows the -15-V power supply to track the +15-V power supply. Both are derived from a single secondary fused in both legs by a pair of 2-A slow-blow fuses. The +15-V power supply is an LM317 monolithic voltage regulator incorporating an adjustable resistor network.

The -15-V power supply is a 7912 controlled by an LM301 opamp that senses the difference between the +15-V and -15-V outputs and forces the -15-V output to track the +15-V output. A balance control is provided to trim the -15-V output. The +15-V power supplies are not ground-referenced to the +5-V or +12-V power supplies unless the analog boards are installed in the 224X chassis. All voltage measurements must be referenced to the correct ground. The +15-V power supplies provide 750 mA and are current and thermal protected.

Mains Circuit. The mains circuit for the 224X uses a dual primary transformer with 120-V and 100-V taps (a pair of DPDT switches select the operating voltage: 100, 120, 220, 240 V). This supply is switched on both sides of the line. A primary fuse is provided on the chassis ahead of the RFI filter unit. Fan power is maintained at 120 V by placing the fan across one of the 120-V primaries.

4 PERFORMANCE TESTS AND CALIBRATION

Performance tests are used to check the operation of the unit. Always execute the performance tests before proceeding to calibration.

4.1 Test Equipment Required

The following equipment is needed for performance tests and calibrations:

1. Variable ac voltage source with isolation transformer, voltmeter, and ammeter
2. Digital voltmeter (DVM)
3. Dual trace oscilloscope with >60-MHz bandwidth
4. Audio band low distortion sine wave generator with a 20-dBm maximum output
5. Harmonic distortion analyzer with level meter
6. Noise meter
7. High-quality music source
8. Lexicon-compatible footswitch and footpedal
9. Headphone amplifier and headphones
10. Cables and dip clips
11. Extender card (Lexicon no. 750-01850)
12. 1.15-ohm, 30-W resistor.

See Figs. 4.2 and 4.3 at the end of this section for interior views of the 224X mainframe.

4.2 System Tests

Diagnostics, nonvolatile storage, and listening tests. When the 224X is turned on or reset, it runs a series of diagnostic programs.

1. Make sure that the unit passes all power-up diagnostics. Repeat the diagnostics several times.
2. Store a control-head setting in a register, then leave the unit off for a while (>1 min). Turn on the unit and make sure that the setting is restored. Check to see that the contents of the register are unchanged by calling the register.
3. Using various signals from signal generators and music sources, listen carefully to all programs and variations. Make sure that there are no excess or unusual noises, birdies, or intermittents.
4. Make sure that moving or gently shaking the control head or mainframe does not affect its output.

Visual Inspection. Inspect the 224X and control head for obvious signs of physical damage. If possible, compare it with a unit operating properly. Remove the front panel of the unit and make the following checks:

1. The protective shield should be in place for the power switch and wiring.

2. The fuse ratings should be as follows:

Primary

100/120 V	3 AG 3 A slow blow
220/240 V	3 AG 1.5 A slow blow

Secondary*

F1, F2	+15 Vac	2 A slow blow
F3, F4	+12 Vac	3 A slow blow
F5	+10 Vac	2 A slow blow
F6, F7	+5 Vac	15 A slow blow
F8	-5 Vdc	2.5 A slow blow

*Schematic fuse no.

- The ac voltage changeover switches should be set correctly (see Sec. 1 of Owner's Manual).
- All jacks, pots, and switches should operate smoothly.
- XLR connectors should be secure.
- There should be no loose screws.
- All ribbon cables and connectors should be secure.
- All ICs should be securely in their sockets.
- There should be no parts missing.

4.3 Power Supplies (PS1, PS2, and PS3)

The nominal and operating line voltages for the 224X are as follows:

<u>Nominal (Vac)</u>	<u>Operating (Vac)</u>
100	90-105
120	108-126
220	198-231
240	216-252

Measure all power supplies. Compare voltages to Table 4.1. All power supplies must provide the correct voltage; if they do not, repair and/or calibration is required.

Table 4.1. Test Point Locations for Power Supplies.

Number	Supply (Vdc)	Limits (Vdc)	Location description	Adjustment Location
1	+5	4.85 to 5.15	SBC module; U16, pin 16: left front-most IC; left front IC pin; verify left LED lit on NVS module	R7 on PS3
2	-5	-4.75 to 5.25	SBC module; J72 at the rear and right of U16**	*
3	+12	11.4 to 12.6	SBC module; R8 front lead; 2.7-Kohm, 1/4-W resistor, left of U15; verify center LED lit on NVS module	*
4	-12	-11.4 to 12.6	SBC module; R4 front lead; 270-ohm, 1/2-W resistor, right of U15	*
5	+15	14.75 to 15.25	AIN module; +15 = test point, ground to test point	R6 on PS2
6	-15	-14.75 to 15.25	AIN module; -15 = test point, ground to test point	R5 on PS2
7	+7	6.3 to 7.7	AIN module; +7 = test point, ground to test point	*
8	-7	-6.3 to 7.7	AIN module; -7 = test point, ground to test point	*
9	+7	6.3 to 7.7	AOUT module; +7 = test point, ground to test point	*
10	-7	-6.3 to 7.7	AOUT module; -7 = test point, ground to test point	*
11	10 Vac	8 to 14 Vac	Power molex connector on Transition module	*

*No adjustment.

** To access this test point, turn off the 224X and loosen the SBC module from its backplane connector, connect a test lead to the J72 test point, and then reinsert the SBC module into the backplane.

4.4 Analog Tests

Apply a 1-kHz, +12-dBm sine wave from the oscillator to both input channels. (This signal is used for standard input tests.) Run the zero-delay test program.

4.4.1 Input sensitivity

Adjust the signal generator output for +8 dBm. Advance input gain adjust pots R1 and R2 until the onset of clipping is reached (the gain pots should be close to their full clockwise rotation). Next, set the signal generator for +18 dBm and rotate the input gain pots counterclockwise (CCW) to just below the clipping level. This should be close to full CCW rotation. Design tolerances allow most machines to operate within a +7- to +22-dBm input range, even though the device is rated at +8 to +18 dBm. Adjust R1 and R2 so that the "0 dB" LED headroom indicator just goes off when the signal generator output is set at +12 dBm.

4.4.2 Output sensitivity

Check each channel for audio output. Make sure that each output can deliver +8- to +18-dBm output into a 600-ohm load by varying the output gain potentiometers; then set each output gain for an output level of +12 dBm.

4.4.3 Frequency response

Check frequency response for each output channel and compare with the frequency response curves in Fig. 4.1. Make sure that the frequency response is within system specification: ± 1.5 dB from 20 Hz to 15 kHz; ± 0.5 dB from 20 Hz to 12 kHz.

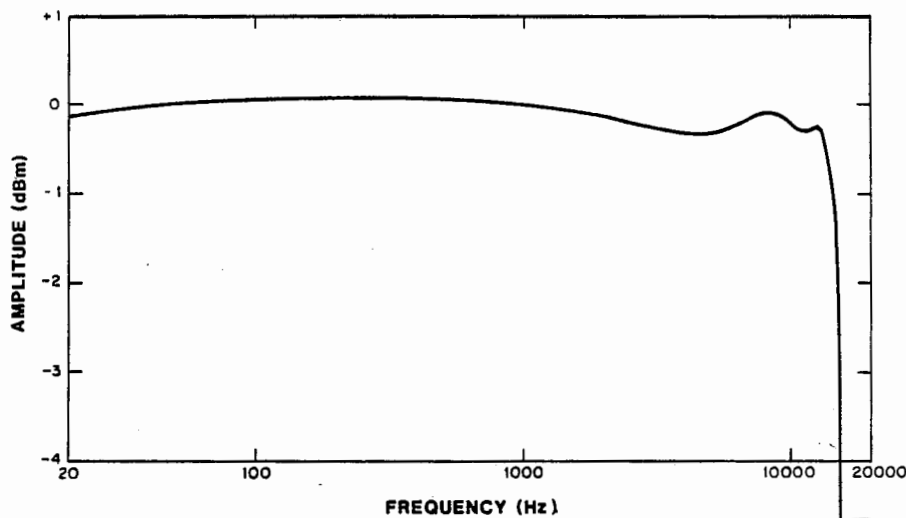


Fig. 4.1. Typical Overall System Frequency Response.

4.4.4 Total harmonic distortion (THD) and noise

THD and noise measured at 1 kHz and 10 kHz should be as follows:

Signal Level (dBm)	Frequency (kHz)	THD and Noise (%)
+12	1	<0.04
0	10	<0.50

4.4.5 Noise floor/signal-to-noise ratio

With 600-ohm loads connected to both inputs and outputs, the noise level at each output should be within the following limits:

	Wide Band (20 Hz to 20 kHz)	A-Weighted
Noise floor	<-68 dBm	<-80 dBm
Ratio relative to +12 dBm	>80 dB	>92 dB

4.4.6 Channel separation

On the left channel input, remove the signal source and connect a 600-ohm load. Apply the standard input test signal to the right channel input. Residual signal measured in output channels A and D should be 60 dB or more below +12 dBm (-48 dBm or below).

Next, remove the input signal to the right channel input and connect a 600-ohm load. Apply the standard input test signal to the left channel input. Residual signal measured in output channels B and C should be -48 dBm or below.

4.5 Calibration

Before proceeding with calibration procedures, the performance tests (Secs. 4.1 to 4.4) should be carried out to determine whether adjustments are necessary.

4.5.1 +5-V current foldback adjustment

1. Turn off the 224X and remove the NVS module from its slot (labeled OPT.).
2. Clip a 1.15-ohm 30-W resistor onto pins 1 and 3, and attach a DVM's minus lead to pin 1 and its positive lead to pin 3. Insert the extender board into the OPT. slot.
3. Turn on the unit. Adjust R11 on power supply module 3 until the meter reads approximately 3.00 V (2.8 to 3.2 V).
4. Remove the extender board and replace the NVS module.

Caution: The 1.15-ohm resistor must handle at least 30 W; it may become hot during use.

4.5.2 Phase-lock loop calibration

1. Connect channel 1 of the oscilloscope to U27 pin 3 and synchronize on channel 1.
2. Connect channel 2 of the oscilloscope to U27 pin 8 (clip on the end of R10, 1K resistor).
3. Adjust the variable capacitor C12 for a loop control voltage, U27 pin 8, of about 3.8 V.

4.5.3 Input offset calibration

Power down and carefully remove U23. Connect a jumper from pin 6 of the U23 socket (hook onto CR21 cathode) to a quiet ground (for example, low side of C73). Power up and connect one channel of the oscilloscope to U19 pin 1 (CH1L) and the other channel to U26 pin 21 (MSB of A/D word). Synchronize on CH1L (positive going) and set the oscilloscope to 5 us/DIV and 2 V/DIV on each channel. Offset (R90) should now be adjusted so that the MSB (most-significant bit) dithers between 1 and 0 with equal intensity when viewed on the oscilloscope.

Power down, remove jumper, and replace U23. This procedure ensures that the analog-to-digital converter (ADC) responds with code 1000 0000 0000 or 0111 1111 1111 for a true 0-V input. This will ensure good gain step matching during output conversion.

4.5.4 Output offset calibration

Set up mainframe with the AOUT module on an extender card. Apply a 1-kHz, +12-dBm sine wave from the oscillator to both input channels. Run the zero-delay test program. Adjust the input level potentiometers (R1 and R2) on the AIN module so that the overload indicator in the headroom display is just turned off. Measure the level of the output of Channel A. Adjust the Channel A output level potentiometer (R119) so that the output level is +12 dBm. Measure the distortion of the Channel A output. Adjust R4 for minimum distortion.

4.5.5 Output filter calibration

Each of the four output filters has three null adjustments to compensate for component sensitivity. The nulling procedure is done by removing the signals to the audio inputs and then injecting signals at various frequencies at certain points. The following procedure outlined for channel A should be executed on all four channels.

1. Apply a 22.153-kHz signal with peak amplitude of 0.5 V to the high side of R43 and adjust R19 for a null at node R20/R21.
2. Apply a 19.200-kHz signal with peak amplitude of 0.5 V to node R20/R21 and adjust R25 for a null at node R26/R27.
3. Apply a 35.850-kHz signal with peak amplitude of 0.5 V to node R26/R27 and adjust R31 for a null at node R32/R33.
4. If nulls can't be achieved, carefully examine all component values of the stage in question.

Repeat for channels B, C, and D. Refer to schematics for corresponding test points.

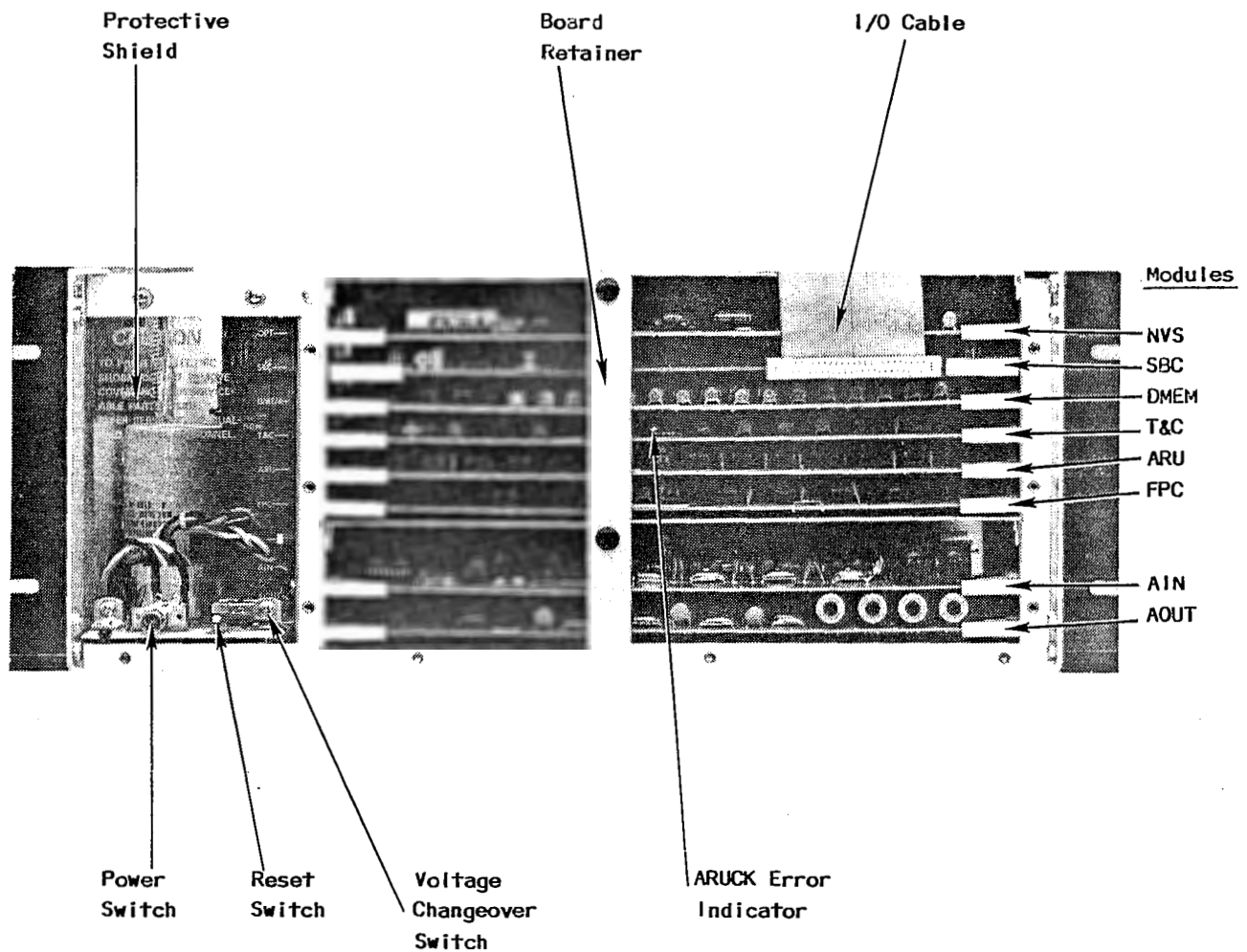


Fig. 4.2. 224X Mainframe Interior -- Front View.

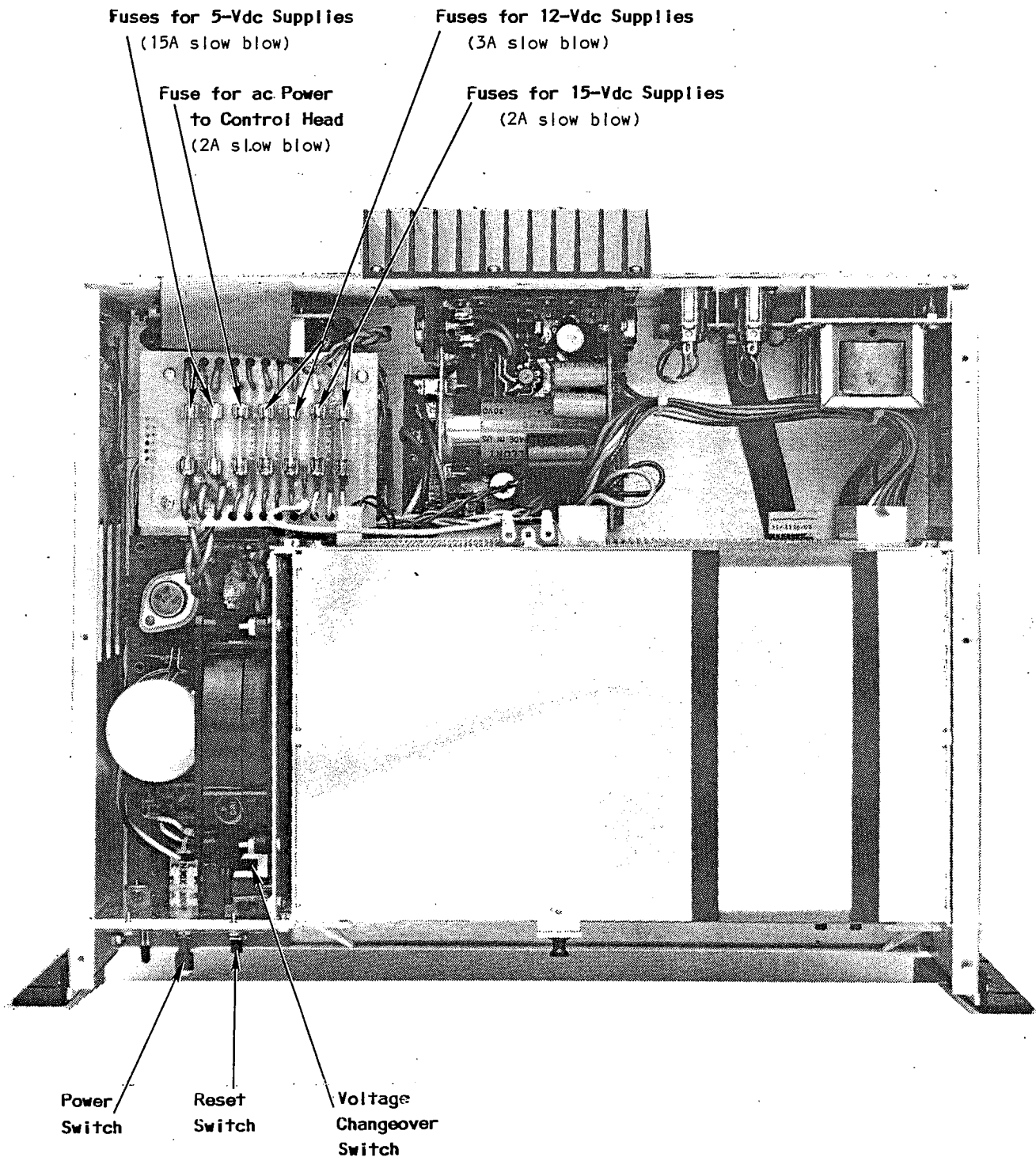


Fig. 4.3. 224X Mainframe Interior -- Top View.

5 TROUBLESHOOTING

5.1 Overall Approach

A thorough visual inspection of the 224X and control head is good general troubleshooting practice. Check for any obvious component failures, such as burnt or overheated parts or damaged PC board traces. There should be no loose ICs, connectors, or cables. Observe whether the malfunction is intermittent, heat related, or shock related.

Figure 5.1 shows a flow chart for troubleshooting problems in the Model 224X. As can be seen, the power-up software diagnostics are an important tool in the troubleshooting process. One of the first things to do is to see whether the unit can run the power-up diagnostics. If these diagnostics do not run, the problem probably is in the power supply module, the SBC module, or the remote control head. (Note that the power-up diagnostics will run with only these three modules in the unit.) First look at the power supplies, then check the cable to the control head. If the problem is with the control head or the connecting cable, and the mainframe is functional, the unit will eventually run one of the reverberation/effects programs (after running the power-up diagnostics), with the remote control head disconnected.

If the unit can run the power-up diagnostics, the diagnostic error codes it produces often supply some information about the nature of the malfunction. Note that the power-up diagnostics test only the digital signal processor (DSP) circuitry, that is, the NVS, SBC, DMEM, T&C, and ARU modules. Although the power-up diagnostics are thorough, they do not test these modules completely. Also, with noise-related or intermittent problems, the power-up diagnostics may not catch an error. Thus, if the unit passes power-up diagnostics, but the reverberation/effects programs sound bad, the DSP may have problems that are not diagnosed by the power-up diagnostics, or the analog conversion system, that is, the FPC, AIN, and AOUT modules, may have problems.

Furthermore, it is possible for the reverberation/effects programs to sound OK and the unit to fail power-up diagnostics. Possibly, the diagnostic hardware or a low-order bit may be malfunctioning. In any case, these kinds of problems should be pursued.

The analog conversion system can be tested on its own through a self-test procedure. Self-test is accomplished by removing the DSP, NVS, DMEM, T&C, and ARU modules. The SBC module is retained to generate a clock signal. If the unit works in the self-test mode, the problem is likely to lie with the DSP.

The 224X has several other diagnostic programs available to help when troubleshooting problems. These programs are called by pushing one of the eight program pushbuttons at the beginning of the power-up diagnostics. The diagnostic programs are listed in Table 5.1.

The first diagnostic programs to run are a maximum delay (0.5-sec) program (diagnostic program 8) and a zero delay program (diagnostic program 7).

These delay programs are similar, except that the maximum delay program uses the DMEM module and the zero delay program does not. Thus, if the unit works in the zero delay program, but not in the maximum delay program, the problem is probably in the DMEM module. These two programs also help in exercising the digital-to-analog interface that is not tested in the self-test mode and in diagnosing where a problem is in the analog conversion system. (Note that the self-test mode does not run at the correct sampling rate when the SBC module is used to generate the clock signal.)

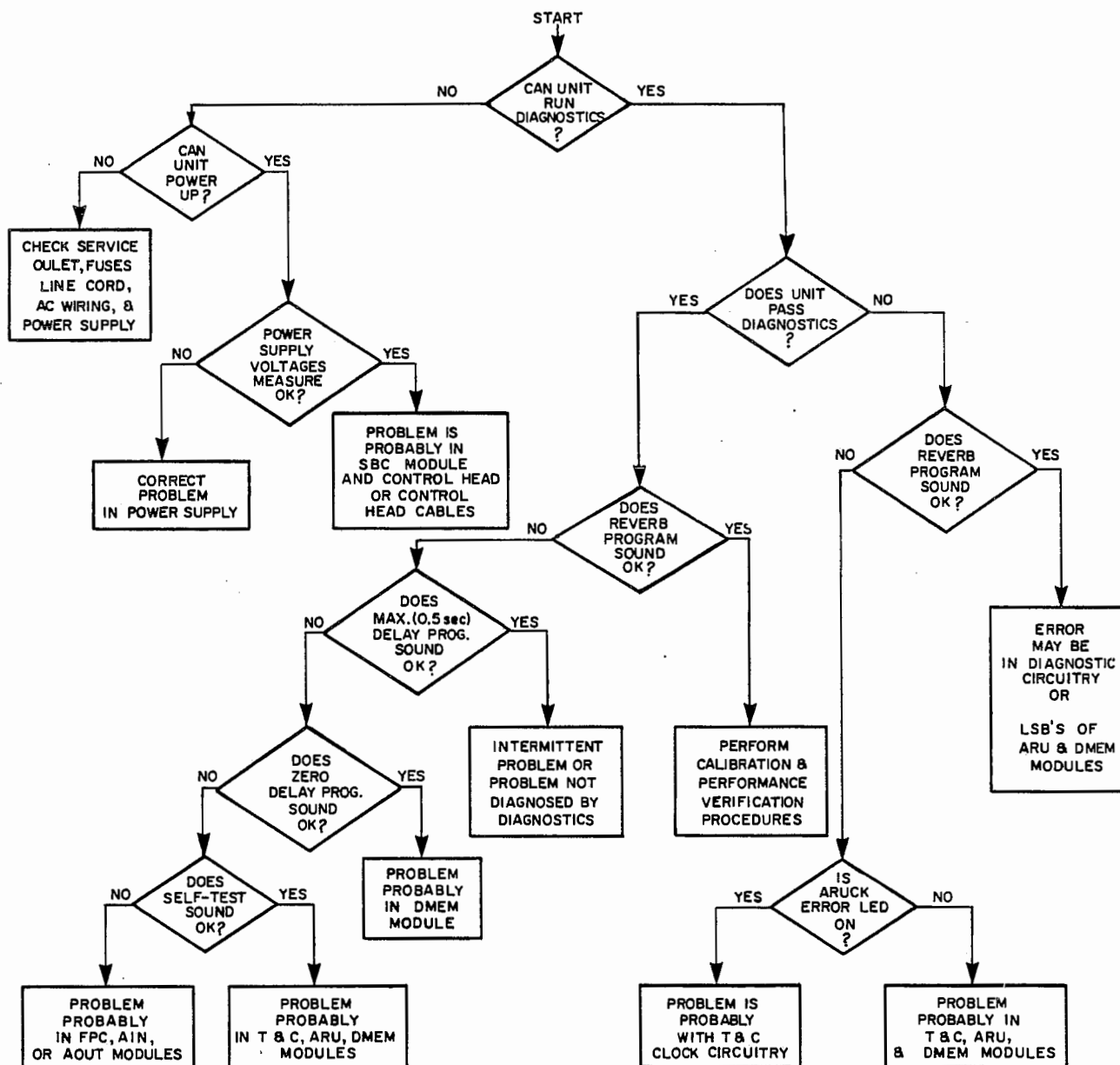


Fig. 5.1. Troubleshooting Approach.

Table 5.1. Diagnostic Programs.

Program*	Description
1 RESTART	restarts self-testing power-up diagnostics and returns to normal operation
2 QUICK EXIT	exits directly to normal operation
3 ARU SIGNAT	generates signature analysis signals; for use by service personnel to test the ARU module
4 ARU TEST	runs a quick test of the ARU module and returns to menu
5 NVS STROBE	generates analysis signals; to test the NVS module
6 FPC SIGNAT	generates signature analysis signals; to test the FPC module
7 ZERO DELAY	loads a 0-second delay-line program for setting input and output levels; Left input passes to outputs A and D, and Right input passes to outputs B and C
8 MAX DELAY	loads a 0.5-second delay-line program for setting input and output levels; Left input passes to outputs A and D, and Right input passes to outputs B and C

*Programs 3, 4, 5, 6, and 7 are not implemented in V8.0 software.

The ARU test (diagnostic 4) exercises the ARU module more thoroughly than the power-up diagnostics and can be tried when the unit passes power-up diagnostics but the ARU is still suspected to be bad. The NVS strobe program (diagnostic 5) continuously sends signals to the NVS module so that they can be checked with an oscilloscope. In addition, two signature programs (diagnostics 3 and 6) can be used with signature analyzers for troubleshooting some of the modules. The signature programs are also useful in troubleshooting with oscilloscopes, because they provide simplified and more observable signals to the various modules.

5.2 Power Supplies

If the 224X does not power up, first check the ac line cord for a good connection and the service outlet for power. Next check the power fuse at the rear of the mainframe. If the fuse is blown, replace with an exact replacement fuse: 3AG 3A slow blow for 100/120 Vac operation, 3AG 1.5A slow blow for 220/240 Vac operation.

The 224X mainframe has a 10-Vac unregulated power supply (ac-fused), three pairs of ac-fused regulated power supplies, and one dc-fused regulated power supply. A blown internal fuse generally indicates a problem in the related power supply circuitry, which should be thoroughly checked out.

Table 5.2 lists the power supply fuses. Replacement fuses should always have the correct rating to ensure protection from circuit damage or fire. Table 5.3 shows the location of each of the power supplies.

Table 5.2. Power Supplies and Fuses.

Schematic Fuse No.	Fuse Board Designation	Power Supply Fusing	Fuse Rating
F1, F2	+15 V	+15 Vac Secondary	2 A slow blow
F3, F4	+12 V	+12 Vac Secondary	3 A slow blow
F5	Viso	+10 Vac Secondary	2 A slow blow
F6, F7	+5 V	+5 Vac Secondary	15 A slow blow
F8	-5 Vdc	-5 Vdc	2.5 A slow blow

Table 5.3. Location of Power Supplies.

Power Supply	Output Supply Voltage/Current Rating	Location	
		Rectified/Filtered	Regulated
+15 V	+15 Vdc/750 mA	PS2	PS2
-15 V	-15 Vdc/750 mA	PS2	PS2
+12 V	+12 Vdc/1.25 A	PS3	PS2
-12 V	-12 Vdc/150 mA	PS3	PS3
+10 Viso	+5 Vdc/500 mA	Control Head	Control Head
+5 V	+5 Vdc/10 A	PS1	PS3
-5 V	-5 Vdc/250 mA	PS1	PS3

The first test of any faulty operation is to make sure all power supplies are working properly.

5.3 Power-up Diagnostics

The 224X diagnostic programs test many components of the digital hardware. Although a diagnostic routine may not point out the exact component that has failed, a faulty module or section can usually be isolated. Diagnostic error messages are easier to understand if you are familiar with the overall operation of the 224X and the hexadecimal numbering system.

Table 5.4 shows number conversion between decimal, hexadecimal, and binary numbering systems. Figure 5.2 shows how errors and correct data are displayed on the control head.

Table 5.4. Number System Conversion.

Decimal	Hex	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111
16	10	0001 0000
.	.	.
.	.	.
.	.	.
255	FF	1111 1111

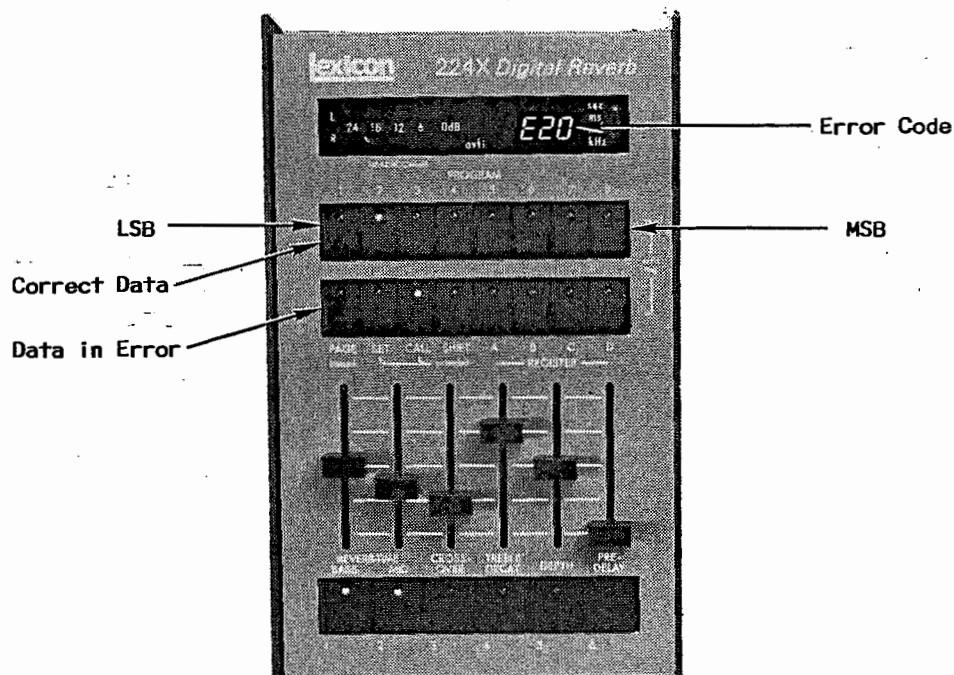


Fig. 5.2. Control Head Error Display.

The 224X diagnostic programs are run whenever the machine is turned on or reset. They can also be run by holding the SHIFT key while pushing CALL and PROGRAM 1 (SHIFT/CALL/PGM 1). The diagnostics make a single pass through all testable features of the machine. If the 224X passes all tests, normal operation begins. If an error is detected, an error message is displayed.

The diagnostic sequence varies for the nonvolatile storage (NVS) module, depending on how the diagnostics are started. When the diagnostics are started by powering up or resetting the 224X, a checksum test is performed on all ROMs on the NVS module, but the nonvolatile RAM is not tested because data can be lost if the power becomes unstable during testing. When the key sequence CALL/SHIFT/PGM 1 is used to start the diagnostics, testing of the nonvolatile RAM on the NVS module is included in the diagnostic program sequence. When the NVS module's nonvolatile RAM is tested, the contents are loaded into the RAM on the SBC module and reloaded to the NVS module when the test is finished. This procedure ensures that registers are not lost as a result of testing.

Caution: Do not reset or power down the unit while the NVS diagnostic is in progress; otherwise contents of the NVS registers may be lost.

During a diagnostic test, or when an error is displayed, all controls and pushbuttons on the control head, except the PROGRAM pushbuttons, are inactive. Briefly pushing PGM 1 allows the unit to go to the next diagnostic test after an error display. Pushing PGM 2 bypasses all further diagnostics and starts normal system operation. Pushing PGM 7 calls a zero delay test program and pushing PGM 8 calls a maximum delay test program.

The diagnostic programs compare the actual data in one part of the machine to the data that should be there if the machine were working perfectly. The diagnostic programs display the expected data pattern (the good data) using the LEDs on the PROGRAM pushbuttons. If different from the expected data pattern, the pattern of the actual data resident in the machine is displayed using the mode/register pushbutton LEDs. Usually an error is displayed when the two data patterns do not agree. The patterns displayed by these LEDs are essential in determining the cause of error. If errors appear, the error message numbers and the lighting patterns of the LEDs should be noted in the order of occurrence when diagnosing or referring a problem to Lexicon service technicians. Table 5.5 is a summary of the 224X error codes, Sec. 5.3.1 describes in detail how to run the programs during troubleshooting, and Sec. 5.3.2 gives in-depth descriptions of the error codes.

Table 5.5. Summary of 224X Error Codes.

Error Type*	Cause of Error
EOX	SBC ROM checksum
E1X	SBC RAM
E2X, E3X, E4X	T&C (may also be DMEM)
E5X - E8X	ARU (may also be T&C or DMEM)
E91, E92, E95, E96	DMEM
H0X, H2X	NVS ROM checksum
H1X	NVS RAM
H2F, H10	NVS card missing

* Where X is from 0 to 9 or from A to F.

5.3.1 Running the diagnostics

The 224X diagnostic programs are run in a particular order. In general, if the diagnostics indicate an error, the source of the first error should be found before much time is spent on any others. The checksum and microprocessor tests come first. Errors here may not disrupt the reverberation or effects programs and are easy to interpret. The remaining diagnostics work by loading the T&C module with a simple program and testing the effects of the program on the machine. The microprocessor must communicate with the T&C module to perform these tests. Failure to communicate usually results in an E23 error, which means that either the timing is incorrect between the T&C module and the microprocessor or a memory chip on the T&C module is faulty. The T&C module has a single LED diagnostic indicator labeled ARUCK ERROR. If this LED lights, the timing signal ARUCK (ARU Clock) is missing, indicating a failure within the clock circuitry on the T&C module. Because the entire machine depends on the T&C module, any problem on this module generally gives errors in other diagnostic tests, even if other sections are not faulty.

Some parts of the tests cannot be easily separated. All tests of the DMEM module require that the ARU module work at least minimally. Most of the microprocessor communication and test circuits for both the T&C and ARU modules are on the DMEM module. A failure here may cause the T&C test to take a long time to complete and may also cause an unusual flashing pattern of the LEDs on the control head. In addition, a short circuit or defective IC on the ARU or FPC modules can cause an error that appears to be on the T&C module. It is sometimes useful to unplug the ARU and FPC modules to see if the error still occurs. The FPC module is not testable by the microprocessor, and the machine should pass all diagnostics with the FPC module unplugged.

If the machine does not appear to work at all, the cable to the control head should be checked first. If no diagnostic errors occur, the machine will eventually run the sonic ambience programs with the control head unplugged. If so, the control head or the cable is faulty. However, it is more likely that a transformer plug in the power supply has come loose, or that a power supply fuse has opened. See Sec. 5.2 for the power supply test points. Failure of the machine to respond with either diagnostics or reverb programs when the power supplies are good probably means that the SBC module is faulty.

5.3.2 Descriptions of error codes

E00 through EOF: SBC ROM checksum test. Each ROM in the SBC module is checked by adding all data bytes; if the data is correct, the sum is 0. If an error is detected, one of the error messages E00 through EOF is displayed, depending on which ROM is faulty. The least-significant hexadecimal digit of the error code represents an encoded nibble (4-bit quantity) binary value. Within that binary value, the bit positions corresponding to a logic 1 identify which chip numbers are faulty. ROM 1 (U23) is indicated by a binary equivalent of 1, ROMs 2 and 3 are both indicated by a binary equivalent of 6, etc., for example:

Error Message	Binary Representation	Faulty ROM*
E01	0001	1
E02	0010	2
E06	0110	2 and 3
EOF	1111	1, 2, 3, and 4
EOA	1010	2 and 4

*ROM 1 = U23, 2 = U25, 3 = U24, and 4 = U26

Note that a checksum error does not always result in a noticeable machine malfunction.

E10 through E13: SBC module RAM test. The RAM on the SBC module is tested using a semirandom pattern, altering all contents. Errors are indicated by display of one of the error messages E10 to E13, depending on the address of the incorrect byte; for example, E10 corresponds to hexadecimal addresses 3C00 to 3CFF and E11 corresponds to addresses 3D00 to 3DFF. The correct pattern is displayed on the Program LEDs, and any incorrect pattern is displayed on the mode/register LEDs. The bit positions corresponding to the elements of the patterns that do not match isolate the location of the error. The error messages and bit patterns are defined as follows:

Error Messages

E10 = RAM addresses 3C00-3CFF
 E11 = RAM addresses 3D00-3DFF

E12 = RAM addresses 3E00-3EFF
 E13 = RAM addresses 3F00-3FFF

Bit Patterns

	LSB*				MSB				
	1	2	3	4	5	6	7	8	
PROGRAM LEDs	0	0	0	0	0	0	0	0	Correct Data
Mode LEDs	0	0	0	0	0	0	0	0	Incorrect Data
Bit No.	0	1	2	3	4	5	6	7	(Bits 0 to 3 are in U36 of the SBC module and bits 4 to 7 are in U37)

*LSB = least significant bit; MSB = most significant bit

The test is accomplished in two passes. First, the memory is loaded with the contents of ROM 1, starting from location zero. This data is complemented twice in two separate passes. The data in RAM is then compared with the original data in ROM. (Data is read from the top down.) The test will stop as soon as any error has been detected. This RAM test is sensitive to addressing errors and data errors.

Control Head Display Test. The panel test outputs data to the control head to light all LEDs (except the 7-segment displays, which display the revision number of the resident software). The test is bypassed if PROGRAM 1 is pushed. The control head display should blank out after the panel test for about 10 to 15 seconds, while the unit is performing diagnostics. Failure of the DMEM and T&C modules to return a ready signal to the SBC module causes the writable control store (WCS) memory test to require more than a minute to complete. If the 224X is not initialized after several minutes, the protect circuitry on the T&C module may be locking the 8080 on the SBC in a hold state. This condition means that the HALT decoding circuitry on the DMEM module or the protect gating on the T&C module is faulty.

E20 through E23: Writable Control Store (WCS) memory test. The WCS (U3, U18, U33, U48, and associated circuits on the T&C module) are tested in the same way as the RAM on the SBC module is tested. The error codes, E20 to E23, are as follows:

Error Message	Hexadecimal Address	Component
E20	4000, 4004 ... 41FC	U49 & associated circuits
E21	4001, 4005 ... 41FD	U33 & associated circuits
E22	4002, 4006 ... 41FE	U18 & associated circuits
E23	4003, 4007 ... 41FF	U3 & associated circuits

Correct data is displayed on the program LEDs, and faulty data is displayed on the mode/register LEDs. Error message E23 could mean that the entire program memory is faulty. If so, or if a problem on the DMEM module prevents data from being properly loaded into the program memory, the memory test will take a long time to complete. Such a delay indicates faults in the SBC bus-decoding checks in the DMEM module or on the T&C module. Two of the memory chips can be swapped to see if different error messages are displayed. Other problems, such as open or shorted address lines, or defective address or data buffers, can also cause a faulty program memory. The diagnostics detect such problems, but require additional steps to isolate them.

E32: 8080 bus test register test. If an error is found when checking the contents of the 8080 bus test register, the 8080 microprocessor in the SBC module cannot communicate with the DSP (consisting of the T&C, ARU, and DMEM modules). If an actual system error occurs, as opposed to a fault in the diagnostic test, it will seriously affect proper operation of further diagnostics as well as the reverberation and effects programs. Short circuits on the T&C or DMEM modules may be the cause. Check to see if the error occurs with the T&C module unplugged. If so, the DMEM module, the Backplane, or the SBC module is faulty. If the reverberation or effects programs seem to be running properly, the bus test register itself (U34 on the DMEM module) is faulty, and the error can be ignored, because it will not interfere with the proper operation of the reverberation and effects programs in the machine.

E40 & E41: Halt and single-step mode tests. Both the halt and single-step mode tests display an error if the return data matches, indicating the halt state cannot be initiated. Because the first program step is continually repeated in the halt state and many tests load known data into the first program step and look for that data at the output of the T&C module, it is necessary to be able to initiate the halt state for subsequent diagnostics to be valid. The halt test works by loading an instruction in the second executable address of the WCS memory. This instruction transfers data from the input of the XREG to the output, and a value loaded into the XREG is checked against the value output. If a halt condition has occurred within the 224X, the values will be different. If no halt has occurred, the values will be identical.

E43: XREG read/write test. The ability of the DSP to read correctly from and write correctly to the transfer register (XREG) on the DMEM module is tested. The WCS memory is first loaded with a NOP and then with an instruction to read the XREG and write to it. If the DSP cannot execute these functions, it means either a faulty XREG itself, or incorrect decoding of the bus control bits of the T&C module. It is important that these errors be corrected before further diagnostics are run, because incorrect XREG contents may invalidate results in all subsequent tests. This error may indicate short circuits or faulty chips at any location along the DAB (digital audio bus). The ARU and FPC modules may be unplugged to be sure they do not contribute to this error. If the same error code is still given, the problem is probably with the transfer registers themselves (U38, U39, U40, and U41 on the DMEM module).

E51 to E7F: ARU register test. All four ARU registers are tested. The registers are composed of four 16-bit memory chips, with each chip organized into four rows of nibbles (4-bit quantities). The first row of chips 1, 2, 3, and 4 constitute the 16 bits of register 1, with chips 1 and 2 containing the higher-order byte of the register and chips 3 and 4 containing the lower-order byte. The second row of the four chips constitutes register 2, the third row register 3, and the fourth row register 4.

Error messages E51 to E5F indicate one or both of the chips comprising the lower-order byte of a register is faulty, E61 to E6F indicate a faulty chip or chips in the higher-order byte of a register, and E71 to E7F indicate chips in both bytes are faulty. The registers are identified by the least-significant hexadecimal digit of the error code. This represents an encoded nibble value. The positions of the bits in a logic 1 state in the binary representation of the nibble value indicate which registers are faulty. For example, the 5 in error code E51 represents a lower-order byte of a register and the 1 corresponds to a binary representation of 0001, indicating an error in register 1. The 9 in E59 corresponds to a binary code of 1001, indicating an error in registers 1 and 4, the 3 in E53 corresponds to 0011, indicating an error in registers 1 and 2, and the 4 in E54 corresponds to 0100, indicating an error in register 3, etc. The following chart shows an example. Incorrect bits within a register are displayed by the mode/register pushbutton LEDs.

ARU register decoding

Error Message	Address Bytes	Binary Code of LSD	Register Number
E67	high	0111	1, 2, and 3
E52	low	0010	2
E69	high	1001	1 and 4
E75	high and low	0101	1 and 3
E7B	high and low	1011	1, 2, and 4

To understand the significance of these diagnostics, examine the following conditions:

1. The multiplier is OK, but the registers are faulty.
 - a. If both high and low bytes are faulty, register addressing is probably invalid. This condition could result in an error in some registers and not in others.
 - b. If a register file chip is faulty, only one byte is affected, and only part of that byte. This may affect all addresses or just one. The data on the LEDs must be checked to determine which chip is actually faulty.

2. The registers are OK, but the multiplier is faulty.

All file addresses are incorrect, and this may propagate errors to areas that are not faulty, so that just replacing the faulty chip does not correct the errors.

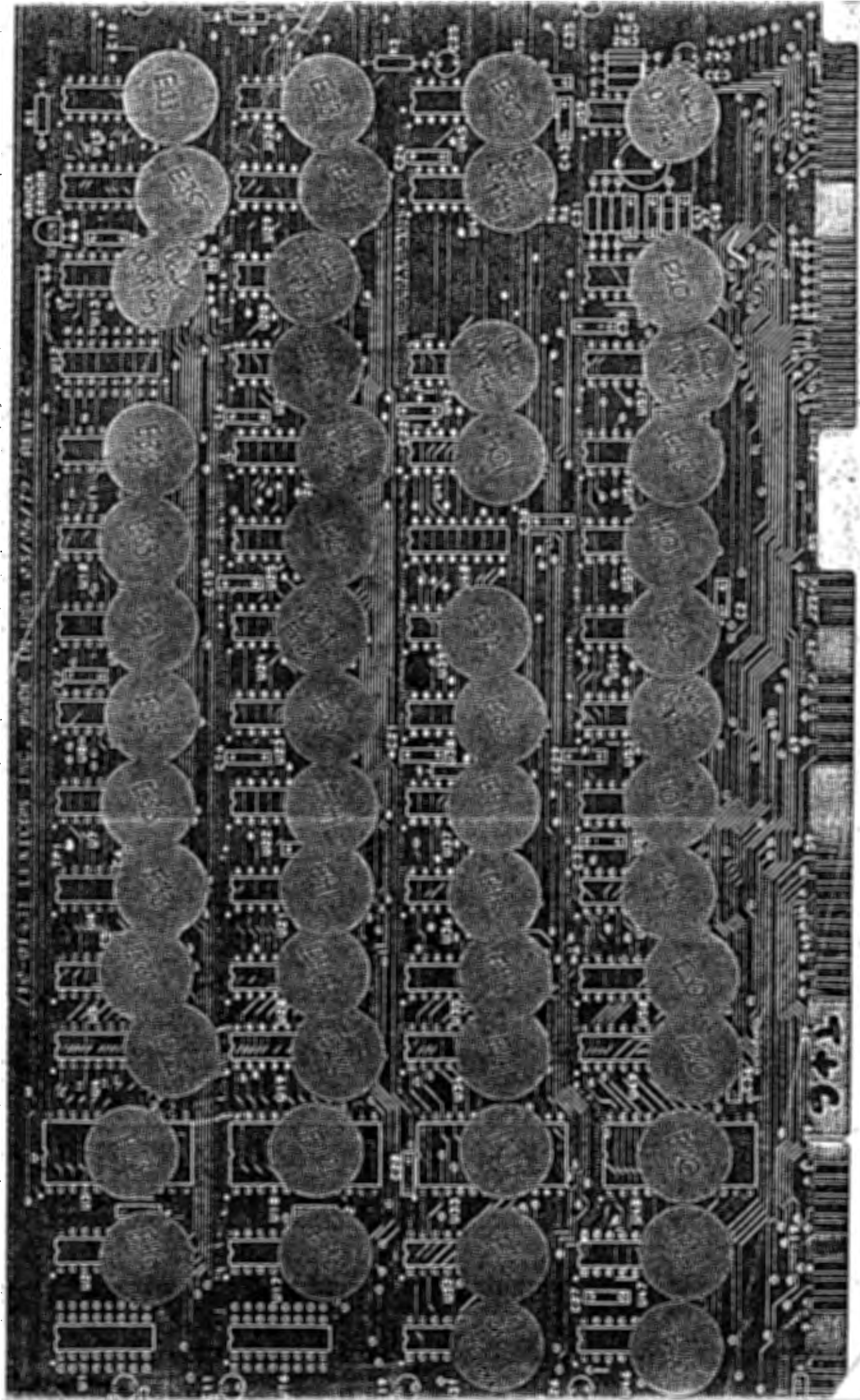
E80 to E8F: Multiplier test. The multiplier test is divided into four parts, each of which displays separate error codes. E81 to E83 indicate an error when the multiply coefficient is $\pm 21/32$. E81 indicates a low-order byte error, E82 a high-order byte error, and E83 indicates both bytes are incorrect. E85 to E87 indicate an error when multiplying by $\pm 42/32$. E89 to E8B indicate an error in the coefficient $\pm 63/64$. Four multiplications are made with each coefficient. The incorrect and correct data are displayed on the pushbutton LEDs in the same way as previous tests. If both the high-order and low-order bytes are incorrect, the data in the low-order byte is displayed. The last two coefficients should set the saturation monostable multivibrator. If not, error code E80 is displayed.

Note that E89 to E8B will test the intermediate address. Therefore, if E89 to E8B is displayed, check U13, U24 to U25, and U38 and U39.

E91, E92, E95, and E96: data memory test. This test makes two passes with complementary data. For this test to be valid, the multiplier must be operating correctly. Error message E91 indicates a low-order byte error and E92 indicates a high-order byte error. If both bytes are incorrect, E91 is displayed, which may mean that the entire memory is faulty, or it may mean that the multiplier or T&C module is faulty. The memory test is sensitive to both data errors and addressing errors, and should detect most problems. If either E91 or E92 are the only errors displayed and only a single LED indicates incorrect data, the problem is most likely to be a faulty memory chip, which is identified by the pushbutton LED. The defective memory chip can be swapped with the lowest-order memory chip (U1). This increases the noise level of the machine by 6 dB, but allows immediate operation if needed. For units with two banks of 16K dynamic RAMs. E91 and E92 refer to bank 1 (U20 to U35), and E95 and E96 refer to bank 2 (U1 to U16).

EA0 to EB3: Additional Multiplier Test. This test is performed when diagnostic program 4 (not available in V8.0) is called. Even error codes indicate a problem with the low byte and the odd error codes indicate a problem with the high byte. The error codes are grouped in fours; in any group, the first two groups use even test data and the second two use odd test data. EA0 to EA3 test a multiply by 1. EA4 to EA7 test a multiply by $1/2$. EA8 to EAB test a multiply by -1 . EAC to EAF test a multiply by $1/4$. EBO to EB3 test a multiply by $5/4$.

If EA0 to EA3 all appear, the coefficients into the 74LS00 (U14, U26 to U28, U40 and U41, and U50 to U53) should be checked. If the high byte is bad with the bad data reading hexadecimal 80 or 7F, check the saturation logic (U42) or the MSB's of the data path through the ARU. If EA0 to EA3 appear, but EA4 to EA7 do not, check the 74LS00's or the intermediate adder (U13, U24, U25, U38, and U39). If EA0 to EA7 appear but EA8 to EAB





CT

1971

1971



do not, check the 74S86's (U5 to U9). EAC to EAF test the shift registers (U3, U4, and U15 to U18) and coefficients. EBO to EB3 test the second adders (U19 to U23).

5.4 Maximum Delay (0.5-sec) Program

The 224X's maximum delay diagnostic program is stereo and can be used as an abbreviated machine test (or for setting output levels). This program can be called during normal operation by the key sequence CALL/SHIFT/PGM 2, or by pressing PGM 8 at the beginning of the power-up diagnostics. The nonadjustable delay provided by this program is approximately half a second. In this program, the left input is passed to outputs A and D; and the right input is passed to outputs B and C.

5.5 Zero Delay Program

The zero delay program is similar to the maximum delay program except that the DMEM module is not used and the delay is zero; that is, the analog inputs are immediately transmitted to the analog outputs. (The digitized signals still pass through the ARU module, however.) This program is called by pushing PGM 7 at the beginning of the power-up diagnostics. In this program, the left input is passed to outputs A and D; and the right input is passed to outputs B and C. The zero delay program is recommended for troubleshooting the analog circuitry and also for setting input and output levels.

5.6 Self-Test Mode

If operating difficulties arise, it is possible to determine proper operation of the AIN, AOUT, and FPC modules by performing a self-test. This test usually isolates problems in the analog conversion subsystem. To perform the self-test, loosen the DMEM, T&C, ARU, and NVS modules. When these modules are removed from the motherboard, a control signal (FPC DBUG) to the FPC module goes HIGH, initiating self-test mode. Do not remove the DMEM, T&C, ARU, and NVS modules when you loosen them; just free them from their edge connectors. If there is no response of any kind in self-test mode, check the power supplies and connections, as well as the clock on the SBC module. Then perform a stage-by-stage investigation from the AIN to the AOUT module to determine where the problem is.

The self-test allows the 224X analog and digitizing subsystems to operate without any intervention from the DSP. This mode digitizes analog information from the left input, and outputs the information to outputs A and B; the right channel input is passed to outputs C and D. Note that the signal routing is different than that in both the maximum delay and zero delay programs. This can sometimes help in determining which channel is bad in a malfunctioning AIN or AOUT module. For example, if the output B is bad in self-test but good in the zero delay program, then the left input channel is bad. If output B is bad in both self-test and zero delay program, then probably the B output channel is bad.

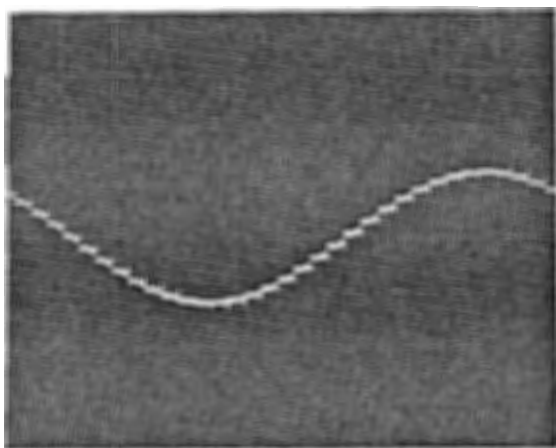
The digital sampling rate in the self-test is only 20 kHz. Therefore, when using the self-test, do NOT use input signals above 10 kHz. The self-test mode samples at only 20 kHz; therefore, it is meant for a preliminary check of the operation of the FPC, AIN, and AOUT modules. If measurements and calibrations are necessary, the zero delay program should be used. The self-test mode can also be exercised at the proper sampling rate of 34.13 kHz by removing the SBC module also and injecting a 3.413-MHz TTL level compatible clock from an external signal generator to pin A28 at the backplane connector for the SBC module. The self-test mode exercises about 95% of the circuitry in the AIN and AOUT modules and about 85% of the circuitry in the FPC module.

5.7 Signature Analysis

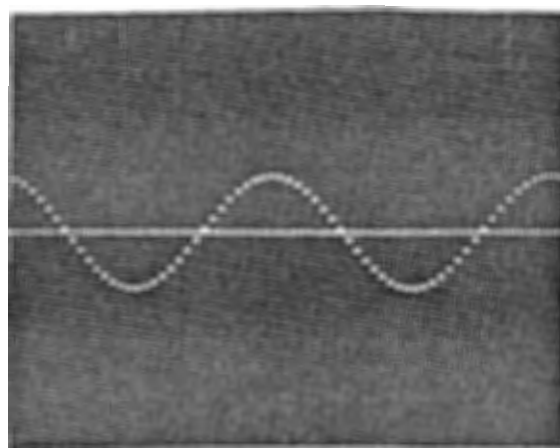
Because the digital signals on the digital modules in the 224X can be quite complex, signature analysis tables have been provided as a useful aid in tracing problems to a malfunctioning node.

Signature analysis is a technique used to troubleshoot electronic logic circuits. A signature analyzer (Hewlett Packard 5004A or equivalent) is connected to the unit being tested and the test program of the unit being tested is started. Long, complex data stream patterns are compressed into a unique 4-segment "signature" that the analyzer will display for each point in the unit being tested as the analyzer probe is moved from point to point. The analyzer requires several signals from the unit under test: the clock signal synchronizes the analyzer and the unit under test; the start and stop signals define the bounds between which the data signal is examined by the analyzer. After the stop signal, the analyzer displays the signature of the data it received. If the signature displayed does not match the corresponding signature given in the table, the circuitry connected to the node is malfunctioning.

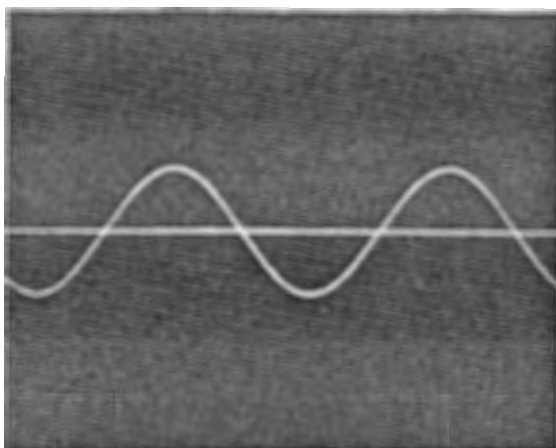
The correct signatures for the various modules in the 224X are summarized in the signature tables that follow. The conditions for taking the signatures are listed with each module and setup. Figure 5.3 shows what some of the waveforms should look like.



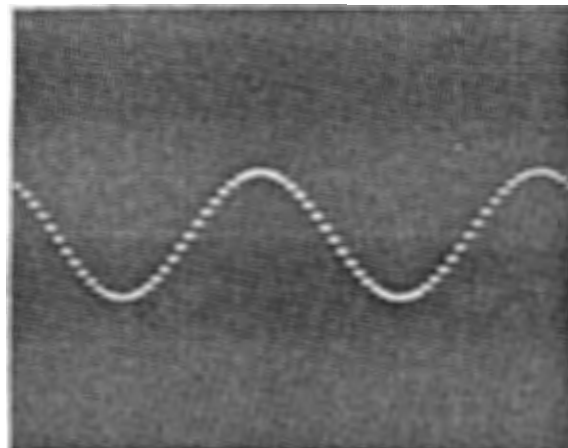
U21 pin 5, AIN board (AGR)
(output of input S/H)
2 V/DIV., 0.1 ms/DIV.
1 kHz input



U22 pin 4, AIN board
(output of channel MUX)
2 V/DIV., 0.2 ms/DIV.
1 kHz input



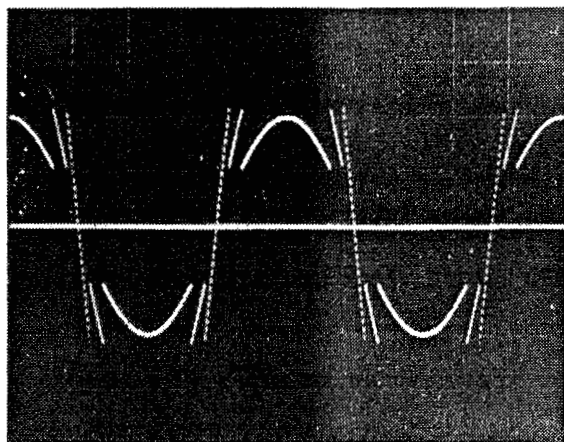
U4 pin 6, AOUT board
(output of output gain range
amplifier)
2 V/DIV., 0.2 ms/DIV.
1 kHz input



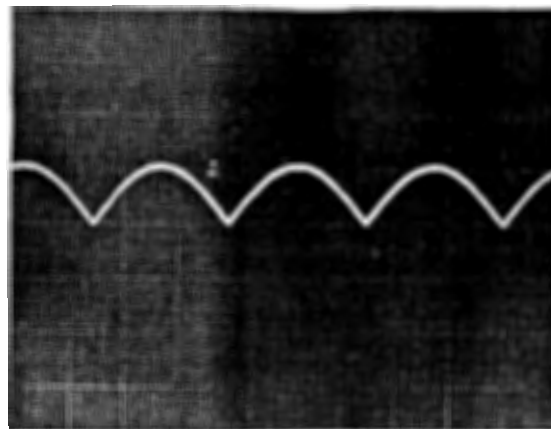
U8 pin 1, AOUT board
(output of output S/H)
2 V/DIV., 0.2 ms/DIV.
1 kHz input

These waveforms are observed with the unit in zero delay program.
Input signal level is +12 dBm to the left input channel;
input level potentiometer is set so that the level indicator
is just beneath overload.

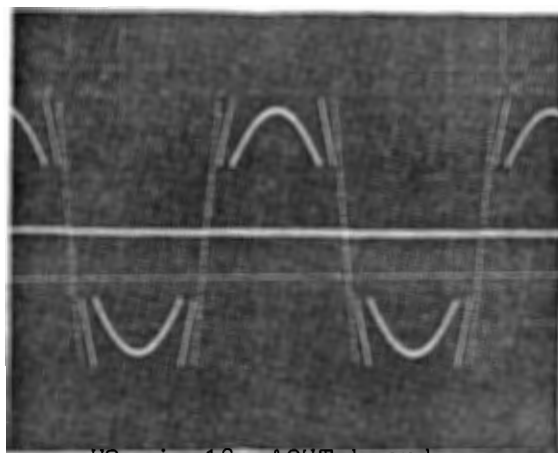
Fig. 5.3. Converter Waveforms.



U23 pin 6, AIN board
(output of input gain range
amplifier)
2 V/DIV., 0.2 ms/DIV.
100 Hz input



U12 pin 7, AIN board
(output of gain range
rectifier)
2 V/DIV., 0.2 ms/DIV.
1 kHz input



U2 pin 18, AOUT board
(output of DAC)
2 V/DIV., 2 ms/DIV.
100 Hz input

These waveforms are observed with the unit in zero delay program. Input signal level is +12 dBm to the left input channel; input level potentiometer is set so that the level indicator is just beneath overload.

Fig. 5.3 (cont'd.). Converter Waveforms.

T & C -- Version 8.2.1

SETUP = in Diagnostic Program 3 (ARU Signatures).
Refer to Schematic #060-02475.

NOTE: Blue control head should display EOA.

START = RESET U19 pin 9 
STOP = RESET U19 pin 9 
CLOCK = DAB RSTB/ U20 pin 6 
+5V = FP54
GROUND = 0000

U1	1	-	16	FP54
	2	-	15	-
	3	-	14	3U9F
	4	-	13	0000
	5	-	12	0000
	6	-	11	-
	7	FP54	10	40A5
	8	0000	9	FP54

U2	1	0000	24	FP54
	2	07P6	23	45FF
	3	FP4C	22	F344
	4	03UA	21	95CP
	5	3U9U	20	A36H
	6	F5AA	19	3U9F
	7	A0A8	18	0000
	8	53PH	17	0000
	9	028H	16	FP54
	10	-	15	0000
	11	0000	14	0000
	12	0000	13	-

U3	1	07P6	20	FP54
	2	FP4C	19	-
	3	03UA	18	-
	4	3U9U	17	-
	5	F5AA	16	-
	6	A0A8	15	-
	7	53PH	14	-
	8	028H	13	-
	9	FP54	12	-
	10	0000	11	FP54

U4	1	FP54	16	FP54
	2	-	15	-
	3	07P6	14	03U3
	4	FP4C	13	6725
	5	03UA	12	01UH
	6	3U9U	11	1UFU
	7	0000	10	0000
	8	0000	9	0000

U5	1	FP54	16	FP54
	2	-	15	-
	3	F5AA	14	P2H5
	4	A0A8	13	H054
	5	53PH	12	29U6
	6	028H	11	8146
	7	0000	10	0000
	8	0000	9	0000

U10	1	FP54	16	FP54
	2	-	15	-
	3	-	14	-
	4	1UFU	13	8146
	5	H054	12	-
	6	8146	11	F1C3
	7	8146	10	-
	8	0000	9	-

T & C -- Version 8.2.1 (Cont'd.)

U11	1	FP54	16	FP54	U14	1	-	16	FP54
	2	-	15	-		2	-	15	-
	3	-	14	-		3	-	14	45FF
	4	01UH	13	29U6		4	-	13	F344
	5	P2H5	12	-		5	-	12	95CP
	6	2PU6	11	FPAA		6	-	11	A36H
	7	29U6	10	-		7	FP54	10	FP54
	8	0000	9	-		8	0000	9	FP54
U15	1	0000	24	FP54	U16	1	C5A9	20	FP54
	2	C5A9	23	45FF		2	909P	19	-
	3	909P	22	F344		3	0000	18	-
	4	0000	21	95CP		4	0000	17	-
	5	0000	20	A36H		5	8C99	16	-
	6	8C99	19	3U9F		6	0H10	15	-
	7	UH10	18	0000		7	0000	14	-
	8	0000	17	0000		8	7302	13	-
	9	7302	16	FP54		9	FP54	12	-
	10	-	15	0000		10	0000	11	FP54
	11	0000	14	0000					
	12	0000	13	-					
U17	1	FP54	16	FP54	U18	1	FP54	16	FP54
	2	-	15	-		2	-	15	-
	3	C5A9	14	U3AA		3	8C99	14	PFC2
	4	909P	13	484U		4	0H10	13	2UU6
	5	0000	12	0000		5	0000	12	0000
	6	0000	11	0000		6	7302	11	3981
	7	0000	10	0000		7	0000	10	0000
	8	0000	9	0000		8	0000	9	0000
U19	1	-	20	FP54	U20	1	-	16	FP54
	2	0000	19	FP54		2	-	15	-
	3	0000	18	FP54		3	-	14	FP54
	4	PFC2	17	6725		4	-	13	-
	5	PFC2	16	6725		5	-	12	-
	6	2UU6	15	03U3		6	-	11	-
	7	2UU6	14	03U3		7	-	10	FP54
	8	FP54	13	3981		8	0000	9	9FF0
	9	672A	12	3981					
	10	0000	11	-					

T & C -- Version 8.2.1 (Cont'd.)

U28	1	FP54	16	FP54
	2	-	15	0000
	3	3U9F	14	-
	4	3U9F	13	-
	5	-	12	-
	6	0000	11	-
	7	0000	10	0000
	8	0000	9	0000

U29	1	0000	24	FP54
	2	4496	23	45FF
	3	P265	22	F344
	4	2P14	21	95CP
	5	F71H	20	A36H
	6	C10H	19	3U9F
	7	A80H	18	0000
	8	CA09	17	0000
	9	909P	16	FP54
	10	-	15	0000
	11	0000	14	0000
	12	0000	13	-

U30	1	4496	20	FP54
	2	P265	19	-
	3	2P14	18	-
	4	F71H	17	-
	5	C10H	16	-
	6	A80H	15	-
	7	CA09	14	-
	8	909P	13	-
	9	FP54	12	-
	10	0000	11	FP54

U31	1	0000	20	FP54
	2	7132	19	A24C
	3	P265	18	4496
	4	F71H	17	2P14
	5	638P	16	970A
	6	H406	15	71U8
	7	A80H	14	C10H
	8	909P	13	CA09
	9	484U	12	HH04
	10	0000	11	-

U32	1	A24C	14	FP54
	2	A24C	13	7132
	3	A24C	12	7132
	4	970A	11	7132
	5	970A	10	638P
	6	970A	9	638P
	7	0000	8	638P

U33	1	-	14	FP54
	2	-	13	-
	3	484U	12	-
	4	861C	11	-
	5	-	10	-
	6	-	9	-
	7	0000	8	-

U34	1	861C	14	FP54
	2	830C	13	3981
	3	0000	12	-
	4	861C	11	-
	5	HH05	10	-
	6	0001	9	U7H5
	7	0000	8	-

U42	1	FP54	16	FP54
	2	-	15	0000
	3	45FF	14	-
	4	45FF	13	A36H
	5	-	12	A36H
	6	F344	11	-
	7	F344	10	95CP
	8	0000	9	95CP

T & C -- Version 8.2.1 (Cont'd.)

U43	1	0000	24	FP54
	2	0616	23	45FF
	3	7P28	22	F344
	4	F237	21	95CP
	5	CA0C	20	A36H
	6	0616	19	3U9F
	7	7P28	18	0000
	8	F184	17	0000
	9	1C45	16	FP54
	10	-	15	0000
	11	0000	14	0000
	12	0000	13	-

U44	1	0616	20	FP54
	2	7P28	19	-
	3	F237	18	-
	4	CA0C	17	-
	5	0616	16	-
	6	7P28	15	-
	7	F184	14	-
	8	1C45	13	-
	9	FP54	12	-
	10	0000	11	FP54

U45	1	0000	20	FP54
	2	3U14	19	830C
	3	7P28	18	0616
	4	CA0C	17	F237
	5	HH05	16	611C
	6	3U14	15	830C
	7	7P28	14	0616
	8	1C45	13	F184
	9	8HA2	12	F9CF
	10	0000	11	-

U46	1	FP54	16	FP54
	2	FP54	15	-
	3	-	14	FP54
	4	FP54	13	-
	5	FP54	12	FP54
	6	FP54	11	FP54
	7	FP54	10	FP54
	8	0000	9	FP54

U47	1	0000	16	FP54
	2	U3AA	15	-
	3	484U	14	71U8
	4	-	13	H406
	5	484U	12	-
	6	U3AA	11	-
	7	CCP5	10	-
	8	0000	9	-

U48	1	U3AA	14	FP54
	2	-	13	-
	3	-	12	6725
	4	861C	11	-
	5	-	10	-
	6	-	9	-
	7	0000	8	-

U49	1	-	14	FP54
	2	861C	13	F9CF
	3	3U14	12	-
	4	861C	11	861C
	5	-	10	FP54
	6	-	9	8HA2
	7	0000	8	55C6

ARU -- Version 8.2.1 -- no feedback

SETUP = Diagnostic Program 3 (ARU Signatures).
Refer to Schematic #060-01318.

NOTE: Blue control head should display EOA.

START = RESET/ extender card pin 16 
STOP = XFERCK U43 pin 11 
CLOCK = ARUCK U10 pin 11 
+5V = 29F3
GROUND = 0000

U2	1	-	14	29F3
	2	-	13	C356
	3	-	12	9A95
	4	-	11	C356
	5	9A95	10	9A95
	6	C356	9	C8P7
	7	0000	8	9124

U3	1	29F3	16	29F3
	2	F494	15	C77H
	3	0000	14	0C0P
	4	0000	13	-
	5	-	12	-
	6	-	11	0000
	7	-	10	8FC4
	8	0000	9	29F3

U4	1	29F3	16	29F3
	2	U69F	15	8575
	3	46PU	14	8653
	4	0000	13	-
	5	-	12	-
	6	-	11	0000
	7	-	10	8FC4
	8	0000	9	29F3

U5	1	3H61	14	29F3
	2	9A95	13	9A95
	3	A7U4	12	P600
	4	9A4H	11	7F95
	5	9A95	10	9A95
	6	00H8	9	5779
	7	0000	8	FHPF

U6	1	91A4	14	29F3
	2	9A95	13	9A95
	3	0C31	12	CU83
	4	HF27	11	2516
	5	9A95	10	9A95
	6	46C2	9	0AU0
	7	0000	8	9065

U7	1	697U	14	29F3
	2	9A95	13	9A95
	3	U3PA	12	PHC1
	4	7U69	11	7724
	5	9A95	10	9A95
	6	P5UF	9	5FF8
	7	0000	8	F65H

U8	1	9A15	14	29F3
	2	9A95	13	9A95
	3	0080	12	4070
	4	H39H	11	HAP5
	5	9A95	10	9A95
	6	4908	9	8441
	7	0000	8	1PH4

U9	1	A175	14	29F3
	2	9A95	13	9A95
	3	3CP0	12	2272
	4	6750	11	C8P7
	5	9A95	10	9A95
	6	UHF5	9	2272
	7	0000	8	C8P7

ARU -- Version 8.2.1 -- no feedback (Cont'd.)

U10	1	0000	20	29F3	U11	1	0000	20	29F3
	2	2272	19	9A15		2	PHC1	19	91A4
	3	3PA1	18	4P6P		3	A127	18	590F
	4	3PA1	17	HH7P		4	F3H4	17	F20A
	5	2272	16	H39H		5	5FF8	16	HF27
	6	6750	15	8441		6	7U69	15	0AU0
	7	C4P4	14	72F6		7	8497	14	6UA4
	8	38AU	13	UAA5		8	A8CC	13	0543
	9	A175	12	4070		9	697U	12	CU83
	10	0000	11	0000		10	0000	11	0000
U12	1	29F3	16	29F3	U13	1	4PHU	16	29F3
	2	P600	15	3H61		2	4CU5	15	8595
	3	-	14	-		3	2FP9	14	78P0
	4	C645	13	0087		4	0087	13	H4C6
	5	H4C6	12	4PHU		5	0097	12	A6FH
	6	-	11	-		6	29H3	11	3F61
	7	5779	10	9A4H		7	29F3	10	C645
	8	0000	9	0000		8	0000	9	2FP9
U14	1	8575	14	29F3	U15	1	29F3	16	29F3
	2	90U2	13	90U2		2	4FF4	15	4FF4
	3	78P0	12	C77H		3	3312	14	0U19
	4	F494	11	2FP9		4	8244	13	57U8
	5	90U2	10	90U2		5	PH68	12	6U15
	6	A6FH	9	8653		6	C156	11	0000
	7	0000	8	0097		7	-	10	8FC4
U16	1	29F3	16	29F3	U17	1	29F3	16	29F3
	2	4FF4	15	4FF4		2	UF7F	15	UCC9
	3	3312	14	35FC		3	3312	14	3PF5
	4	5F3P	13	75F4		4	5F3P	13	A574
	5	AC87	12	UF7F		5	9895	12	F494
	6	U7C9	11	0000		6	F4AC	11	0000
	7	-	10	8FC4		7	-	10	8FC4
	8	0000	9	29F3		8	0000	9	29F3
U18	1	29F3	16	29F3	U19	1	6A6H	16	29F3
	2	6U15	15	F9C1		2	00H8	15	FHPF
	3	75UH	14	0FFH		3	H9P3	14	U0U6
	4	1AH1	13	977F		4	1PU7	13	8P4F
	5	HP7A	12	U69F		5	0A55	12	HUPH
	6	8244	11	0000		6	A7U4	11	7F95
	7	-	10	8FC4		7	C356	10	7A3U
	8	0000	9	29F3		8	0000	9	3P92

ARU -- Version 8.2.1 -- no feedback (Cont'd.)

U20	1	2161	16	29F3	U21	1	HCUP	16	29F3
	2	46C2	15	9065		2	P5UF	15	F65H
	3	5941	14	CA1P		3	5938	14	44PA
	4	7819	13	P813		4	94F5	13	CA12
	5	4HCA	12	6HA3		5	A547	12	9152
	6	0C31	11	2516		6	U3PA	11	7724
	7	3P92	10	8AHH		7	F268	10	HPH3
	8	0000	9	F268		8	0000	9	38A5
U22	1	6P49	16	29F3	U23	1	F2A5	16	29F3
	2	4908	15	1PH4		2	UHF5	15	C8P7
	3	1UP4	14	7H7H		3	07F5	14	H491
	4	313U	13	5C0F		4	97CF	13	54H3
	5	091A	12	85A6		5	94U9	12	H491
	6	0080	11	HAP5		6	3CP0	11	C8P7
	7	38A5	10	67P6		7	38A5	10	54H3
	8	0000	9	38A5		8	0000	9	-
U24	1	C4P4	16	29F3	U25	1	8497	16	29F3
	2	0CHA	15	819U		2	3P83	15	2HF9
	3	9PP9	14	9PP9		3	9CF3	14	FUFA
	4	38AU	13	3PA1		4	A8CC	13	F3H4
	5	9321	12	9PP9		5	4380	12	11P7
	6	8767	11	819U		6	8777	11	943H
	7	2FP9	10	3PA1		7	6F4F	10	A127
	8	0000	9	-		8	0000	9	24UH
U26	1	F9C1	14	29F3	U27	1	OFFH	14	29F3
	2	90U2	13	90U2		2	4471	13	4471
	3	FUFA	12	UCC9		3	3P83	12	F9C1
	4	OFFH	11	9CF3		4	UCC9	11	943H
	5	90U2	10	90U2		5	4471	10	4471
	6	4380	9	UF7F		6	2HF9	9	3PF5
	7	0000	8	11P7		7	0000	8	8777
U28	1	8575	14	29F3	U29	1	-	16	29F3
	2	4471	13	4471		2	-	15	-
	3	3F61	12	0C0P		3	-	14	29F3
	4	8653	11	29H3		4	U7C9	13	29F3
	5	4471	10	4471		5	AC87	12	29F3
	6	4CU5	9	C77H		6	PH68	11	0000
	7	0000	8	8595		7	5F3P	10	3312
						8	0000	9	8244

ARU -- Version 8.2.1 -- no feedback (Cont'd.)

U30	1	-	16	29F3	U31	1	-	16	29F3
	2	-	15	-		2	-	15	-
	3	-	14	29F3		3	-	14	29F3
	4	U7C9	13	29F3		4	U7C9	13	29F3
	5	AC87	12	29F3		5	AC87	12	29F3
	6	75UH	11	0000		6	HP7A	11	0000
	7	U7C9	10	AC87		7	5F3P	10	3312
	8	0000	9	C156		8	0000	9	1AH1
U32	1	-	16	29F3	U33	1	0000	16	29F3
	2	-	15	-		2	1PU7	15	0000
	3	-	14	29F3		3	9124	14	7A3U
	4	U7C9	13	29F3		4	1PU7	13	9124
	5	AC87	12	29F3		5	6A6H	12	7A3U
	6	46PU	11	0000		6	9124	11	8P4F
	7	F4AC	10	9895		7	6A6H	10	9124
	8	0000	9	8244		8	0000	9	8P4F
U34	1	0000	16	29F3	U35	1	0000	16	29F3
	2	7819	15	0000		2	94F5	15	0000
	3	9124	14	8AHH		3	9124	14	HPH3
	4	7819	13	9124		4	94F5	13	9124
	5	2161	12	8AHH		5	HCUP	12	HPH3
	6	9124	11	P813		6	9124	11	CA12
	7	2161	10	9124		7	HCUP	10	9124
	8	0000	9	P813		8	0000	9	CA12
U36	1	0000	16	29F3	U37	1	0000	16	29F3
	2	313U	15	0000		2	97CF	15	0000
	3	9124	14	67P6		3	9124	14	54H3
	4	313U	13	9124		4	97CF	13	C8P7
	5	6P49	12	67P6		5	F2A5	12	54H3
	6	9124	11	5C0F		6	9124	11	54H3
	7	6P49	10	9124		7	F2A5	10	C8P7
	8	0000	9	5C0F		8	0000	9	54H3
U38	1	HH7P	16	29F3	U39	1	F20A	16	29F3
	2	968C	15	2H9U		2	3PH5	15	F772
	3	2A87	14	3P2C		3	95C9	14	F1CU
	4	4P6P	13	72F6		4	590F	13	6UA4
	5	45PP	12	1789		5	U2F4	12	1789
	6	2U7H	11	8160		6	8721	11	7P86
	7	24UH	10	UAA5		7	2FP9	10	0543
	8	0000	9	2FP9		8	0000	9	6F4F

ARU -- Version 8.2.1 -- no feedback (Cont'd.)

U40	1	U69F	14	29F3
	2	4471	13	4471
	3	3PH5	12	A574
	4	977F	11	F772
	5	4471	10	4471
	6	7P86	9	F494
	7	0000	8	8721

U41	1	977F	14	29F3
	2	90U2	13	90U2
	3	F1C0	12	A574
	4	3PF5	11	95C9
	5	90U2	10	90U2
	6	1789	9	U69F
	7	0000	8	U2F4

U42	1	54H3	14	29F3
	2	54H3	13	-
	3	0000	12	-
	4	-	11	-
	5	-	10	29F3
	6	-	9	29F3
	7	0000	8	0000

U43	1	2839	20	29F3
	2	-	19	-
	3	97CF	18	6P49
	4	67P6	17	5C0F
	5	-	16	-
	6	-	15	-
	7	54H3	14	HPH3
	8	F2A5	13	313U
	9	-	12	-
	10	0000	11	-

U44	1	2839	20	29F3
	2	-	19	-
	3	94F5	18	2161
	4	8AHH	17	P813
	5	-	16	-
	6	-	15	-
	7	CA12	14	7A3U
	8	HCUP	13	7819
	9	-	12	-
	10	0000	11	-

U45	1	3108	16	29F3
	2	0000	15	-
	3	7A3U	14	HUPH
	4	8P4F	13	U0U6
	5	6A6H	12	H9P3
	6	1PU7	11	0A55
	7	0000	10	0000
	8	0000	9	0000

U46	1	3108	16	29F3
	2	0000	15	-
	3	8AHH	14	6HA3
	4	P813	13	CA1P
	5	2161	12	5941
	6	7819	11	4HCA
	7	0000	10	0000
	8	0000	9	0000

U47	1	3108	16	29F3
	2	0000	15	-
	3	HPH3	14	9152
	4	CA12	13	44PA
	5	HCUP	12	5938
	6	94F5	11	A547
	7	0000	10	0000
	8	0000	9	0000

U48	1	3108	16	29F3
	2	0000	15	-
	3	67P6	14	85A6
	4	5C0F	13	7H7H
	5	6P49	12	1UP4
	6	313U	11	091A
	7	0000	10	0000
	8	0000	9	0000

U49	1	3108	16	29F3
	2	0000	15	-
	3	54H3	14	H491
	4	54H3	13	H491
	5	F2A5	12	07F5
	6	97CF	11	94U9
	7	0000	10	0000
	8	0000	9	0000

ARU — Version 8.2.1 -- no feedback (Cont'd.)

U50	1	UF7F	14	29F3
	2	4471	13	4471
	3	2U7H	12	57U8
	4	6U15	11	8160
	5	4471	10	4471
	6	968C	9	75F4
	7	0000	8	2H9U

U51	1	6U15	14	29F3
	2	90U2	13	90U2
	3	45PP	12	33FC
	4	75F4	11	1789
	5	90U2	10	90U2
	6	2A87	9	57U8
	7	0000	8	3P2C

U52	1	OU19	14	29F3
	2	4471	13	4471
	3	OCHA	12	35FC
	4	4FF4	11	8767
	5	4471	10	4471
	6	819U	9	4FF4
	7	0000	8	819U

U53	1	OU19	14	29F3
	2	90U2	13	90U2
	3	9321	12	4FF4
	4	4FF4	11	9PP9
	5	90U2	10	90U2
	6	9PP9	9	4FF4
	7	0000	8	9PP9

U54	1	HP7A	14	29F3
	2	U7C9	13	C931
	3	0000	12	90U2
	4	29F3	11	6HC2
	5	8244	10	4471
	6	AC87	9	0000
	7	0000	8	29F3

ARU -- Version 8.2.1

SETUP = Diagnostic Program 3 (ARU Signatures).
Refer to Schematic #060-01318.

NOTE: Blue control head should display EOA.

START = RESET/ extender card pin 16
STOP = RESET/ extender card pin 16
CLOCK = ARUCK U10 pin 11
+5V = 3696
GROUND = 0000



U2	1	-	14	3696
	2	-	13	U804
	3	-	12	FP92
	4	-	11	U804
	5	FP92	10	FP92
	6	U804	9	3P2U
	7	0000	8	08C9

U3	1	3696	16	3696
	2	9C33	15	8P47
	3	0000	14	U8A9
	4	0000	13	-
	5	-	12	-
	6	-	11	0000
	7	-	10	4573
	8	0000	9	3696

U4	1	3696	16	3696
	2	343H	15	2149
	3	50U5	14	5465
	4	0000	13	-
	5	-	12	-
	6	-	11	0000
	7	-	10	4573
	8	0000	9	3696

U5	1	85U0	14	3696
	2	FP92	13	FP92
	3	4C62	12	HFUF
	4	H10A	11	126P
	5	FP92	10	FP92
	6	1U98	9	0HP6
	7	0000	8	F374

U6	1	H827	14	3696
	2	FP92	13	FP92
	3	16C5	12	5C3U
	4	876U	11	95AH
	5	FP92	10	FP92
	6	49UH	9	162P
	7	0000	8	H8CF

U7	1	4808	14	3696
	2	FP92	13	FP92
	3	869A	12	3A87
	4	UP8C	11	U415
	5	FP92	10	FP92
	6	3019	9	FCFH
	7	0000	8	055U

U8	1	361P	14	3696
	2	FP92	13	FP92
	3	U88F	12	H021
	4	A1P2	11	1PC3
	5	FP92	10	FP92
	6	6U70	9	3077
	7	0000	8	UPP5

U9	1	C7C3	14	3696
	2	FP92	13	FP92
	3	7921	12	U0CH
	4	6U12	11	3P2U
	5	FP92	10	FP92
	6	A180	9	U0CH
	7	0000	8	3P2U

ARU -- Version 8.2.1 (Cont'd.)

U10	1	0000	20	3696
	2	U0CH	19	361P
	3	CAF1	18	3786
	4	CAF1	17	187U
	5	U0CH	16	A1P2
	6	6U12	15	3077
	7	859P	14	3C55
	8	34HF	13	UCU9
	9	C7C3	12	H021
	10	0000	11	0000

U11	1	0000	20	3696
	2	3A87	19	H827
	3	2PC5	18	PCU4
	4	FF20	17	5565
	5	FCFH	16	876U
	6	UP8C	15	162P
	7	A6AF	14	77P6
	8	FCAC	13	PHF4
	9	4808	12	5C3U
	10	0000	11	0000

U12	1	3696	16	3696
	2	HFUF	15	85U0
	3	-	14	-
	4	P243	13	505C
	5	4077	12	U9AP
	6	-	11	-
	7	OHP6	10	H10A
	8	0000	9	0000

U13	1	U9AP	16	3696
	2	U903	15	6FF1
	3	1P6H	14	3276
	4	505C	13	4077
	5	33UP	12	3P50
	6	5533	11	F27P
	7	3696	10	P243
	8	0000	9	1P6H

U14	1	2149	14	3696
	2	86CP	13	86CP
	3	3276	12	8P47
	4	9C33	11	1P6H
	5	86CP	10	86CP
	6	3P50	9	5465
	7	0000	8	33UP

U15	1	3696	16	3696
	2	QUAH	15	QUAH
	3	3PC5	14	5140
	4	UH8F	13	HU4H
	5	9CPU	12	2C67
	6	F339	11	0000
	7	-	10	4573
	8	0000	9	3696

U16	1	3696	16	3696
	2	QUAH	15	QUAH
	3	3PC5	14	APH6
	4	58H6	13	2344
	5	FC1A	12	UC7F
	6	93FF	11	0000
	7	-	10	4573
	8	0000	9	3696

U17	1	3696	16	3696
	2	UC7F	15	81PA
	3	3PC5	14	567U
	4	58H6	13	9FFP
	5	U5AU	12	9C33
	6	AH79	11	0000
	7	-	10	4573
	8	0000	9	3696

U18	1	3696	16	3696
	2	2C67	15	2PP4
	3	6P40	14	U971
	4	0823	13	33F0
	5	A55A	12	343H
	6	UH8F	11	0000
	7	-	10	4573
	8	0000	9	3696

U19	1	2A65	16	3696
	2	1U98	15	F374
	3	60CU	14	09U9
	4	P885	13	9AH7
	5	5CP3	12	34P3
	6	4C62	11	126P
	7	U804	10	F5F1
	8	0000	9	F7P1

ARU -- Version 8.2.1 (Cont'd.)

U20	1	U9U2	16	3696
	2	49UH	15	H8CF
	3	4815	14	6A06
	4	9136	13	P978
	5	4062	12	011F
	6	16C5	11	95AH
	7	F7P1	10	U058
	8	0000	9	440C

U21	1	PHPF	16	3696
	2	3019	15	055U
	3	0H92	14	8U4F
	4	PC08	13	P9P4
	5	2999	12	CA59
	6	869A	11	U415
	7	440C	10	12U0
	8	0000	9	5F4F

U22	1	411A	16	3696
	2	6U70	15	UPP5
	3	4U40	14	2U18
	4	CHH8	13	C5PH
	5	1918	12	1126
	6	U88F	11	1PC3
	7	5F4F	10	5064
	8	0000	9	UA22

U23	1	6665	16	3696
	2	A180	15	3P2U
	3	4FF8	14	P1UC
	4	0HH7	13	77AH
	5	8PH4	12	P1UC
	6	7921	11	3P2U
	7	UA22	10	77A3
	8	0000	9	-

U24	1	859P	16	3696
	2	6P1P	15	5141
	3	P500	14	P500
	4	34HF	13	CAF1
	5	PU81	12	P500
	6	809C	11	5141
	7	5CF6	10	CAF1
	8	0000	9	-

U25	1	A6AF	16	3696
	2	4H81	15	0C16
	3	FHUC	14	P1P0
	4	FCAC	13	FF20
	5	C6PH	12	AH67
	6	P33P	11	A5A9
	7	9P78	10	2PC5
	8	0000	9	267C

U26	1	2PP4	14	3696
	2	86CP	13	86CP
	3	P1P0	12	81PA
	4	U971	11	FHUC
	5	86CP	10	86CP
	6	C6PH	9	UC7F
	7	0000	8	AH67

U27	1	U971	14	3696
	2	0205	13	0205
	3	4H81	12	2PP4
	4	81PA	11	A5A9
	5	0205	10	0205
	6	0C16	9	567U
	7	0000	8	P33P

U28	1	2149	14	3696
	2	0205	13	0205
	3	F27P	12	U8A9
	4	5465	11	5533
	5	0205	10	0205
	6	U903	9	8P47
	7	0000	8	6FF1

U33	1	000P	16	3696
	2	P885	15	0000
	3	08C9	14	F5F1
	4	P885	13	08C9
	5	2A65	12	F5F1
	6	08C9	11	9AH7
	7	2A6H	10	08C9
	8	0000	9	9AHU

ARU -- Version 8.2.1 (Cont'd.)

U34	1	000P	16	3696
	2	9136	15	0000
	3	08C9	14	U058
	4	9136	13	08C9
	5	U9U2	12	U05P
	6	08C9	11	P978
	7	U9U0	10	08C9
	8	0000	9	P978

U35	1	000P	16	3696
	2	PC08	15	0000
	3	08C9	14	12U0
	4	PC02	13	08C9
	5	PHPF	12	12UF
	6	08C9	11	P9P4
	7	PHPA	10	08C9
	8	0000	9	P9PA

U36	1	U00P	16	3696
	2	CHH8	15	0000
	3	08C9	14	5064
	4	CHHP	13	08C9
	5	411A	12	5062
	6	08C9	11	C5PH
	7	4110	10	08C9
	8	0000	9	C5P9

U37	1	U00P	16	3696
	2	0HH7	15	0000
	3	08C9	14	77A3
	4	0HHH	13	3P2U
	5	6665	12	77A3
	6	08C9	11	77AH
	7	666C	10	3P2U
	8	0000	9	77A3

U38	1	187U	16	3696
	2	9740	15	1A74
	3	29UF	14	82P8
	4	3786	13	3C55
	5	8476	12	C2A0
	6	958C	11	H721
	7	267C	10	UCU9
	8	0000	9	5CF6

U39	1	5565	16	3696
	2	496H	15	47U5
	3	F27F	14	PP67
	4	PCU4	13	77P6
	5	124C	12	9AU6
	6	P7H2	11	P94A
	7	1P6H	10	PHF4
	8	0000	9	9P78

U40	1	343H	14	3696
	2	0205	13	0205
	3	496H	12	9FFP
	4	33F0	11	47U5
	5	0205	10	0205
	6	P94A	9	9C33
	7	0000	8	P7H2

U41	1	33F0	14	3696
	2	86CP	13	86CP
	3	PP67	12	9FFP
	4	567U	11	F27F
	5	86CP	10	86CP
	6	9AU6	9	343H
	7	0000	8	124C

U42	1	77A3	14	3696
	2	77AH	13	-
	3	000P	12	-
	4	-	11	-
	5	-	10	3696
	6	-	9	3696
	7	-	8	0000

U43	1	25C5	20	3696
	2	-	19	-
	3	0HHH	18	4110
	4	5062	17	C5P9
	5	-	16	-
	6	-	15	-
	7	77A3	14	12UF
	8	666C	13	CHHP
	9	-	12	-
	10	0000	11	-

ARU -- Version 8.2.1 (Cont'd.)

U44	1	25C5	20	3696
	2	-	19	-
	3	PC02	18	U9U0
	4	U05P	17	P978
	5	-	16	-
	6	-	15	-
	7	P9PA	14	F5F1
	8	PHPA	13	9136
	9	-	12	-
	10	0000	11	-

U45	1	8658	16	3696
	2	0000	15	-
	3	F5F1	14	34P3
	4	9AHU	13	09U9
	5	2A6H	12	60CU
	6	P885	11	5CP3
	7	0000	10	0000
	8	0000	9	0000

U46	1	8658	16	3696
	2	0000	15	-
	3	U05P	14	011F
	4	P978	13	6A06
	5	U9U0	12	4815
	6	9136	11	4062
	7	0000	10	0000
	8	0000	9	0000

U47	1	8658	16	3696
	2	0000	15	-
	3	12UF	14	CA59
	4	P9PA	13	8U4F
	5	PHPA	12	0H92
	6	PC02	11	2999
	7	0000	10	0000
	8	0000	9	0000

U48	1	8658	16	3696
	2	0000	15	-
	3	5062	14	1126
	4	C5P9	13	2U18
	5	4110	12	4U40
	6	CHHP	11	1918
	7	0000	10	0000
	8	0000	9	0000

U49	1	8658	16	3696
	2	0000	15	-
	3	77A3	14	P1UC
	4	77A3	13	P1UC
	5	666C	12	4FF8
	6	0HHH	11	8PH4
	7	0000	10	0000
	8	0000	9	0000

U50	1	UC7F	14	3696
	2	0205	13	0205
	3	958C	12	HU4H
	4	2C67	11	H721
	5	0205	10	0205
	6	9740	9	2344
	7	0000	8	1A74

U51	1	2C67	14	3696
	2	86CP	13	86CP
	3	8476	12	APH6
	4	2344	11	C2A0
	5	86CP	10	86CP
	6	29UF	9	HU4H
	7	0000	8	82P8

U52	1	5140	14	3696
	2	0205	13	0205
	3	6P1P	12	APH6
	4	0UAH	11	809C
	5	0205	10	0205
	6	5141	9	0UAH
	7	0000	8	5141

U53	1	5140	14	3696
	2	86CP	13	86CP
	3	PU81	12	0UAH
	4	0UAH	11	P500
	5	86CP	10	86CP
	6	P500	9	0UAH
	7	0000	8	P500

ARU -- Version 8.2.1 (Cont'd.)

U54	1	A55A	14	3696
	2	93FF	13	C028
	3	0000	12	86CP
	4	3696	11	3493
	5	UH8F	10	0205
	6	FC1A	9	0000
	7	0000	8	3696

DMEM -- Version 8.2.1

SETUP = Diagnostic Program 3 (ARU Signatures).
 Lift U65 pin 13 and jumper to U65 pin 1.
 Refer to Schematic #060-02512.

NOTE: Blue control head should display E0A.

START = MSB of CPC; U65 pin 8 
 STOP = MSB of CPC; U65 pin 8 
 CLOCK = RESET/ U58A pin 1 
 +5V = 826P
 GROUND = 0000

U1-	1	-	16	0000
U16,	2	-	15	-
U20-	3	-	14	-
U35	4	-	13	0AU1
	5	UH56	12	A6F7
	6	P279	11	FU8U
	7	861U	10	44U7
	8	-	9	-

U17	1	-	16	-
	2	0AU1	15	0AU1
	3	A6F7	14	A6F7
	4	UH56	13	UH56
	5	FU8U	12	FU8U
	6	P279	11	P279
	7	44U7	10	44U7
	8	861U	9	861U

U18	1	0000	16	826P
	2	FU8U	15	0000
	3	1F7P	14	8HU0
	4	FU8U	13	9237
	5	44U7	12	-
	6	6633	11	-
	7	44U7	10	74P1
	8	0000	9	0AU1

U36	1	0000	16	826P
	2	UH56	15	0000
	3	UF4C	14	A6F7
	4	UH56	13	6266
	5	861U	12	A6F7
	6	5439	11	P279
	7	861U	10	440H
	8	0000	9	P279

U48	1	826P	20	826P
	2	826P	19	826P
	3	-	18	-
	4	0000	17	0000
	5	-	16	-
	6	0000	15	0000
	7	-	14	-
	8	0000	13	0000
	9	-	12	-
	10	0000	11	0000

U49	1	5439	16	826P
	2	0000	15	0000
	3	2A1F	14	A206
	4	UF4C	13	440H
	5	7P25	12	C133
	6	0000	11	0000
	7	826P	10	6266
	8	0000	9	9241

U50	1	6633	16	826P
	2	0000	15	826P
	3	3319	14	7C47
	4	1F7P	13	74P1
	5	8P3U	12	C25F
	6	0000	11	0000
	7	9241	10	9237
	8	0000	9	A077

U51	1	-	14	826P
	2	0000	13	C133
	3	7P25	12	0000
	4	2A1F	11	8P3U
	5	A206	10	3319
	6	C133	9	7C47
	7	00U0	8	C25F

DMEM -- Version 8.2.1 (Cont'd.)

U62	1	826P	20	826P
	2	0000	19	826P
	3	-	18	-
	4	0000	17	826P
	5	-	16	-
	6	0000	15	0000
	7	-	14	-
	8	0000	13	0000
	9	-	12	-
	10	0000	11	0000

U64	1	44U7	16	826P
	2	0000	15	0000
	3	19H6	14	HP66
	4	FU8U	13	0AU1
	5	5H21	12	U81P
	6	826P	11	0000
	7	10F0	10	8HU0
	8	0000	9	FPHA

U63	1	861U	16	826P
	2	0000	15	0000
	3	19H6	14	HP66
	4	UH56	13	P279
	5	5H21	12	U81P
	6	0000	11	0000
	7	A077	10	A6F7
	8	0000	9	10F0

U65	1	C25F	14	826P
	2	0000	13	C25F
	3	5H21	12	0000
	4	0000	11	5H21
	5	HP66	10	19H6
	6	U81P	9	HP66
	7	0000	8	U81P

FPC -- Version 8.2.1

SETUP = Diagnostic Program 6 (FPC Signatures).

Lift pin 11 of SAR IC (U26) on AIN module and jumper to +5V.

Refer to Schematic #060-01320.

NOTE: Blue control head should display EOF.

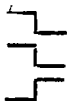
START = RESET U4 pin 2

STOP = RESET U4 pin 2

CLOCK = FPCCLK U4 pin 5

+5V = 96F6

GROUND = 0000



U1	1	96F6	16	96F6
	2	0000	15	-
	3	0000	14	1686
	4	96F6	13	388A
	5	0000	12	AP4F
	6	0000	11	-
	7	80F0	10	80F0
	8	0000	9	C869

U2	1	96F6	16	96F6
	2	0000	15	80F0
	3	0000	14	-
	4	96F6	13	H48C
	5	0000	12	-
	6	96F6	11	-
	7	96F6	10	388A
	8	0000	9	C869

U3	1	96F6	16	96F6
	2	-	15	96F6
	3	5151	14	AP4F
	4	0000	13	UC7P
	5	96F6	12	0000
	6	-	11	96F6
	7	-	10	5F99
	8	0000	9	-

U4	1	0000	16	96F6
	2	4C63	15	0000
	3	96F6	14	5151
	4	4C63	13	H30A
	5	0000	12	5151
	6	-	11	96F6
	7	0000	10	H30A
	8	0000	9	96F6

U5	1	AU07	14	96F6
	2	P11U	13	1H0P
	3	-	12	96F6
	4	1C7C	11	-
	5	CHC5	10	-
	6	96F6	9	2712
	7	0000	8	H30A

U6	1	AU07	16	96F6
	2	P11U	15	0000
	3	1H0P	14	0000
	4	1358	13	0000
	5	1C7C	12	38UP
	6	CHC5	11	7FU6
	7	26HC	10	72A7
	8	0000	9	6HU5

U7	1	4C63	16	96F6
	2	0000	15	0000
	3	-	14	1H0P
	4	-	13	P11U
	5	-	12	AU07
	6	-	11	0000
	7	96F6	10	2712
	8	-	9	96F6

U8	1	4C63	16	96F6
	2	0000	15	2712
	3	-	14	1C7C
	4	-	13	CHC5
	5	-	12	26HC
	6	-	11	1358
	7	96F6	10	96F6
	8	-	9	-

FPC -- Version 8.2.1 (Cont'd.)

U12	1	0000	14	96F6	U13	1	-	14	96F6
	2	-	13	1686		2	-	13	-
	3	96F6	12	H48C		3	-	12	-
	4	-	11	UC7P		4	-	11	-
	5	-	10	87H6		5	-	10	1A65
	6	-	9	C869		6	-	9	4325
	7	0000	8	3UCU		7	0000	8	5940
U14	1	-	14	96F6	U15	1	0000	14	96F6
	2	-	13	2PAU		2	-	13	-
	3	C869	12	-		3	5151	12	-
	4	1110	11	388A		4	-	11	1110
	5	HP96	10	-		5	-	10	87H6
	6	5940	9	-		6	-	9	2PAU
	7	0000	8	-		7	0000	8	C869
U16	1	96F6	16	96F6	U17	1	AU07	14	96F6
	2	0000	15	-		2	-	13	P11U
	3	96F6	14	-		3	-	12	77H9
	4	96F6	13	-		4	-	11	-
	5	96F6	12	36UA		5	-	10	-
	6	-	11	-		6	-	9	6HU5
	7	36UA	10	36UA		7	0000	8	UC33
	8	0000	9	UC33					
U18	1	96F6	16	96F6	U23	1	96F6	16	96F6
	2	-	15	3953		2	-	15	1A65
	3	6C8P	14	-		3	428H	14	4325
	4	41HA	13	72A7		4	9U57	13	2U46
	5	38UP	12	7FU6		5	P3A1	12	6211
	6	-	11	A8CH		6	A12F	11	0000
	7	41HA	10	3P7C		7	010H	10	3UCU
	8	0000	9	0000		8	0000	9	2PAU
U24	1	96F6	16	96F6	U25	1	96F6	20	96F6
	2	-	15	010H		2	96F6	19	96F6
	3	HHHA	14	52A5		3	-	18	-
	4	PA30	13	2513		4	96F6	17	96F6
	5	P3A1	12	76CC		5	-	16	-
	6	H44C	11	0000		6	96F6	15	96F6
	7	010H	10	3UCU		7	-	14	-
	8	0000	9	2PAU		8	36UA	13	96F6
						9	-	12	-
						10	0000	11	96F6

FPC -- Version 8.2.1 (Cont'd.)

U26	1	96F6	20	96F6
	2	7C6U	19	96F6
	3	-	18	-
	4	96F6	17	0000
	5	-	16	-
	6	96F6	15	0000
	7	-	14	-
	8	96F6	13	0000
	9	-	12	-
	10	0000	11	HC53

U27	1	96F6	16	96F6
	2	-	15	HC53
	3	AH63	14	0000
	4	AH63	13	0000
	5	AH63	12	0000
	6	AH63	11	0000
	7	7C6U	10	36UA
	8	0000	9	6HU5

U28	1	96F6	16	96F6
	2	-	15	7C6U
	3	AH63	14	96F6
	4	CA1F	13	96F6
	5	735H	12	96F6
	6	0A75	11	0000
	7	96F6	10	36UA
	8	0000	9	6HU5

U34	1	96F6	16	96F6
	2	-	15	010H
	3	HHHA	14	52A5
	4	PA30	13	2513
	5	P3A1	12	76CC
	6	H44C	11	0000
	7	010H	10	3UCU
	8	0000	9	2PAU

U35	1	96F6	16	96F6
	2	-	15	010H
	3	HHHA	14	-
	4	PA30	13	-
	5	P3A1	12	-
	6	H44C	11	0000
	7	-	10	3UCU
	8	0000	9	2PAU

U36	1	5151	20	96F6
	2	HHHA	19	H44C
	3	-	18	-
	4	-	17	-
	5	PA30	16	P3A1
	6	P3A1	15	PA30
	7	-	14	-
	8	-	13	-
	9	H44C	12	HHHA
	10	0000	11	0000

U37	1	5151	20	96F6
	2	428H	19	H44C
	3	-	18	-
	4	-	17	-
	5	9U57	16	P3A1
	6	P3A1	15	PA30
	7	-	14	-
	8	-	13	-
	9	A12F	12	HHHA
	10	0000	11	0000

U38	1	96F6	16	96F6
	2	-	15	96F6
	3	2550	14	96F6
	4	POC4	13	96F6
	5	H808	12	96F6
	6	9U1U	11	0000
	7	96F6	10	36UA
	8	0000	9	6HU5

FPC -- Version 8.2.1 (Cont'd.)

U39	1	96F6	16	96F6	U40	1	96F6	16	96F6
	2	-	15	96F6		2	0000	15	-
	3	57UH	14	96F6		3	CCC4	14	HHHA
	4	2PP1	13	96F6		4	HHHA	13	A12F
	5	81F2	12	36UA		5	3PAP	12	9U57
	6	9426	11	0000		6	9U57	11	7567
	7	0000	10	36UA		7	0000	10	0000
	8	0000	9	6HU5		8	0000	9	5151
U41	1	96F6	16	96F6	U42	1	000U	16	96F6
	2	0000	15	-		2	779C	15	5F99
	3	HHHA	14	779C		3	P11U	14	F543
	4	A12F	13	49P0		4	C3U7	13	77H9
	5	9U57	12	6HUP		5	49P0	12	43PH
	6	7567	11	F543		6	P11U	11	6HUP
	7	0000	10	0000		7	3833	10	77H9
	8	0000	9	C869		8	0000	9	0276
U43	1	96F6	16	96F6					
	2	0000	15	HP96					
	3	0000	14	584H					
	4	0000	13	9P3F					
	5	96F6	12	-					
	6	96F6	11	-					
	7	1110	10	96F6					
	8	0000	9	C869					

6 SCHEMATICS AND ASSEMBLY DRAWINGS

The following schematics and assembly drawings are contained in this section in the order listed:

<u>Title</u>	<u>Lexicon Drawing No.</u>
Single Board Computer Schematic	
Single Board Computer Assembly	
Arithmetic Unit Board Schematic	060-01318
Arithmetic Unit Board Assembly	030-02735
Floating Point Converter Board Schematic	060-01320
Floating Point Converter Board Assembly	030-01420
Analog Input Board Schematic	060-01321
Analog Input Board Assembly	030-02733
Analog Output Board Schematic	060-01322
Analog Output Board Assembly	030-02734
Power Supplies Schematic	060-01324
Power Supply #1 Assembly	030-01423
Power Supply #2 Assembly	030-01424
Power Supply #3 Assembly	030-01425
Motherboard Schematic	060-01360
Motherboard Assembly	030-01428
Memory Expansion Board Schematic	060-02273
Memory Expansion Board Assembly	080-02281
Timing and Control Board Schematic	060-02475
Timing and Control Board Assembly	030-02481
Data Memory Board Schematic	060-02512
Data Memory Board Assembly	030-02516
Data Memory Board Block Diagram	
Fuse Board Schematic (see Power Supply Schematic)	
Fuse Board Assembly	030-02647
Output Transformer Board Schematic	060-01359
Output Transformer Board Assembly	030-02769
Transition Board Assembly	030-01426
Remote Control Head (Panel) Schematic (Logic and Display)	060-01323
Remote Control Head Assembly	080-01757

Power Supply Module Assembly	080-01611
Power Transformer Assembly	080-01650
Chassis-1 Assembly	080-01662
Chassis-2 Assembly	080-01676
Chassis-3 Assembly	080-01845
LARC Schematic	060-03534
LARC Display Board Assembly	080-03397
LARC Electronics Board Assembly	080-03403
LARC Panel Board Assembly	080-03409
LARC Transition Board Schematic	060-03576
LARC Transition Board Assembly	080-03447